

Graph-Aware Deep Reinforcement Learning for Adaptive Capacity Planning in Distributed Personalized Regenerative Medicine Manufacturing Networks

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Abstract

Autologous cell therapy and other personalized regenerative medicine (PRM) products are manufactured on demand across distributed clinics. Each product is perishable, identity-bound and non-fungible, and useful only while its patient remains clinically eligible, coupling capacity, inventory, patient-indexed logistics, and patient outcomes. We formulate this planning problem as a constrained graph Markov decision process and develop an *advantage-filtered residual graph convolutional network-deep deterministic policy gradient* controller (AFR-GCN-DDPG). It learns bounded corrections to a transparent two-period look-ahead policy while preserving identity and feasibility. In the routing-primary formal study, five training seeds were evaluated on four scenarios with 100 paired holdout replications per seed and scenario. AFR-GCN-DDPG reduced the total weighted objective by 0.658% relative to routing MDL-2 (95% interval $[-0.721, -0.585]\%$) and by 0.321% relative to parameter-matched flat DDPG (95% interval $[-0.413, -0.240]\%$),

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while improving completion service, manufacturing eligibility, and patient loss. Savings were driven primarily by lower patient-loss, expiry, and shortage costs, partly offset by higher specimen-transfer cost. A matched three-seed TD3 development ablation also favored the graph controller over MDL-2 and flat TD3. However, DDPG and TD3 final checkpoints were statistically indistinguishable from frozen pretraining, and a prospective paired-advantage DDPG experiment failed its online-gain gate. Frozen action-geometry screens found that useful local headroom concentrated in projected integer-lot specimen routing rather than continuous reagent and capacity channels. The evidence supports an offline-pretrained, graph-aware residual actor–critic policy, but not an independent endpoint benefit from the tested online updates.

Keywords: Graph convolutional networks, Deep reinforcement learning, Actor–critic methods, Capacity planning, Distributed manufacturing, Personalized regenerative medicine, Perishable inventory, Transshipment

1. Introduction

Distributed make-to-order manufacturing networks are increasingly important in engineering systems where products are individualized, production capacity is scarce, and service outcomes depend on time-sensitive sequential decisions. Personalized regenerative medicine (PRM), and autologous cell therapy in particular, is a demanding instance of this class. Each therapy is manufactured from an individual patient’s own biological material, so production can begin only when that patient’s specimen, an available bioreactor, and sufficient consumables are simultaneously present, and the finished product must be returned to the same patient within a limited viability window.

11 Because the product is patient-specific, it cannot be pre-produced, pooled, or
12 substituted, and the opportunity to buffer against uncertainty with inventory
13 is fundamentally limited [1, 2]. Three properties therefore distinguish PRM
14 capacity planning from conventional make-to-order or inventory control. The
15 product is *perishable*, subject to a hard shelf-life and vein-to-vein deadline; it
16 is *identity-bound*, tied one-to-one to a single patient and non-fungible across
17 patients; and its demand is driven by *patient health*, since a patient whose
18 condition deteriorates while waiting may become clinically ineligible before
19 treatment can be delivered. These are not marginal concerns. Autologous
20 CAR-T manufacturing has a turnaround measured in weeks, an estimated 4–
21 7% of patients never receive their product because of manufacturing failure,
22 and in multiple myeloma roughly one quarter of patients die while waitlisted
23 for a BCMA-directed CAR-T production slot [3]. Discrete-event simulation
24 links longer waits directly to survival, with predicted one-year mortality in
25 large B-cell lymphoma rising from about 36% to 76% as vein-to-vein time
26 increases from one to nine months [4]. With the CAR-T market projected to
27 reach tens of billions of US dollars by the early 2030s [5], delivering perish-
28 able, patient-specific product before patients deteriorate is a consequential
29 operational challenge.

30 These properties make capacity planning a coupled, network-level deci-
31 sion problem rather than a collection of independent local ones. Locating
32 production near patients through distributed or point-of-care manufactur-
33 ing shortens fulfillment time, but it also creates spatial dependencies. A
34 shortage at one clinic can sometimes be relieved by relocating idle bioreactor
35 capacity or transshipping consumables from connected facilities, and each

36 such action changes the future state of the whole network [6, 7]. Identity
 37 binding constrains ownership and use, not physical location: reagents and
 38 idle bioreactor capacity are fungible, whereas each specimen lot and finished
 39 product remain attached to one named patient. An intact waiting specimen
 40 may therefore be routed to a qualified manufacturing facility without being
 41 pooled, split, substituted, or reassigned, and the resulting product must re-
 42 turn to that patient’s collection or infusion site. Coordination must account
 43 for this patient-indexed logistics pathway as well as fungible-resource shar-
 44 ing. The state of such a system is naturally relational, since each facility is a
 45 node whose value depends on its neighbors’ queues and inventories and on the
 46 links along which shortages, slack, specimens, and disruptions propagate. A
 47 flat feature vector encodes local conditions but discards this topology, which
 48 motivates a graph-structured state representation as a matter of substance
 49 rather than convenience.

50 Reinforcement learning (RL) is a natural fit because the problem is se-
 51 quential. An action taken in one period alters future inventories, capacity,
 52 waiting specimens, and service outcomes. Continuous-control actor–critic
 53 methods, including deep deterministic policy gradient (DDPG) [8], twin de-
 54 layed DDPG (TD3) [9], soft actor–critic (SAC) [10], and proximal policy op-
 55 timization (PPO) [11], are standard tools for such real-valued control. Graph
 56 convolutional networks (GCNs) [12] provide a permutation-equivariant rela-
 57 tional inductive bias for encoding networked state. This machinery is by now
 58 well established, and a substantial literature already couples graph learning
 59 with reinforcement learning for supply-chain and resource-allocation prob-
 60 lems [13, 14, 15, 16, 17]. We therefore do not claim novelty in combining

61 graph representation with reinforcement learning.

62 What that literature does not model is the PRM problem itself. Prior
63 graph-and-RL studies for networked operations concentrate on demand fore-
64 casting, inventory-only ordering, or generic enterprise resource allocation,
65 and they represent demand as fungible, abstract requests [13, 15, 18, 19,
66 20]; none model a product that expires, production that must return to the
67 same patient, or demand generated by patient deterioration while waiting.
68 Reinforcement learning has been applied within cell-therapy manufacturing,
69 but at the level of a single bioprocess rather than a network of clinics [2],
70 and recent evidence cautions that learned policies do not uniformly domi-
71 nate well-tuned operational heuristics, particularly under perishability and
72 non-stationarity [21]. The gap we address is thus a problem gap rather than
73 a method gap, namely perishable, identity-bound, patient-condition-driven
74 capacity planning across a clinic network, together with a decision framework
75 and simulation testbed built specifically for it. To the best of our knowledge
76 no public benchmark exists for this setting, so the environment is itself a
77 contribution.

78 We organize the study around five research questions. **(RQ1)** With
79 patient-indexed specimen routing held admissible across policies, does a
80 graph representation of the clinic network improve coordinated specimen,
81 capacity, reagent, and replenishment decisions over flat-state control, and
82 under which supply-disruption and forecast-error conditions is any improve-
83 ment largest? **(RQ2)** Does the routing-primary graph advantage persist
84 under a matched TD3 backbone ablation, and can any gain be attributed sep-
85 arately to online actor-critic updates rather than frozen advantage-filtered

86 pretraining? **(RQ3)** Does explicitly modeling patient condition and product
 87 expiry change the optimal policy and expose failures of condition-blind plan-
 88 ners? **(RQ4)** Does a condition- and expiry-aware temporal encoder, which
 89 retains memory of deterioration trajectories, improve on a static graph en-
 90 coder? **(RQ5)** Do learned policies outperform strong look-ahead heuristics,
 91 not only weak myopic rules, under supply disruption, demand forecast error,
 92 and patient-condition stress?

93 To study these questions we develop, first, a simulation of a multi-site
 94 autologous cell-therapy network in which patient health conditions, patient-
 95 indexed specimen routing, and product and material expiry are first-class
 96 drivers of demand and feasibility, and second, an advantage-filtered residual
 97 GCN-DDPG (AFR-GCN-DDPG) controller that coordinates specimen rout-
 98 ing, capacity, reagent, and replenishment decisions across the network. AFR
 99 is our shorthand for the complete controller design: a transparent MDL-2 an-
 100 chor, a graph-aware bounded residual actor, evidence-filtered teacher targets,
 101 and independently calibrated deployment safeguards. It does not replace or
 102 rename DDPG. The graph component encodes each facility’s operational and
 103 patient-condition state together with information from connected facilities
 104 into network-aware embeddings. A deterministic policy-gradient actor then
 105 learns a bounded, state-dependent correction to the anchor policy, and a fea-
 106 sibility projection converts the corrected action into implementable controls.
 107 Advantage-filtered distillation retains a nonzero correction target only when
 108 common-random-number look-ahead predicts an improvement over the an-
 109 chor; it is a teacher-data construction and regularization step, not a second al-
 110 gorithm or deployed policy. Checkpoint and residual-scale calibration provide

111 a deployment safeguard. DDPG is the canonical proposed instantiation and
 112 provides continuity with prior PRM capacity-planning work; a parameter-
 113 matched graph-aware TD3 study serves as a development-only backbone ab-
 114 lation, while SAC and PPO remain future common-budget routing-primary
 115 comparisons. We adopt a statistically disciplined evaluation protocol (mul-
 116 tiple random seeds, paired Monte Carlo outcomes, and two-level confidence
 117 intervals), because single-seed, mean-only reporting can make deep reinforce-
 118 ment learning results appear stronger and more stable than they are [22, 23].
 119 Consistent with the clinical stakes of the problem, we report patient-service
 120 outcomes such as eligibility and waiting alongside operating cost, since a
 121 lower average cost can mask clinically unacceptable delay.

122 The main contributions of this study are as follows.

- 123 • We formulate distributed capacity planning for *perishable, identity-*
 124 *bound, patient-condition-driven* PRM manufacturing as a constrained
 125 networked sequential decision problem, in which patient deterioration
 126 and product and material expiry are explicit determinants of demand,
 127 feasibility, and cost.
- 128 • We develop a simulation environment realizing this formulation for a
 129 multi-clinic autologous cell-therapy network, with patient-condition dy-
 130 namics, expiry-aware inventory, patient-indexed and identity-preserving
 131 pre-manufacturing specimen routing, finished-product return, and fungible-
 132 resource sharing. Routing is an operational capability of the model
 133 rather than a claimed methodological contribution.
- 134 • We propose AFR-GCN-DDPG, a two-stage advantage-filtered resid-

135 ual graph controller that learns bounded corrections to a transparent
136 look-ahead anchor, retains the filtered teacher signal during DDPG
137 optimization, and applies independently calibrated deployment safe-
138 guards.

- 139 • We establish a paired multi-seed evaluation protocol that separates
140 policy-versus-anchor value, graph-versus-flat representation value, DDPG-
141 versus-TD3 backbone robustness, and final-versus-frozen online attri-
142 bution. The evidence includes a five-seed formal DDPG confirmation
143 and a matched three-seed TD3 development ablation with two-level
144 bootstrap confidence intervals.
- 145 • We report prospective negative attribution tests rather than folding all
146 learned-policy performance into an online-RL claim. Paired critic su-
147 pervision, legal-action transfer diagnostics, and corrected action-geometry
148 screens identify offline advantage-filtered pretraining and graph repre-
149 sentation as the supported sources of value, while bounding the unsup-
150 ported incremental online-update claim.
- 151 • We draw design implications for intelligent decision support in dis-
152 tributed, patient-specific healthcare manufacturing and related time-
153 critical, make-to-order engineering systems.

154 The remainder of this paper is organized as follows. Section 2 formu-
155 lates the distributed PRM capacity planning problem, extended with patient-
156 condition states and product and material expiry. Section 3 reviews personal-
157 ized regenerative medicine operations, reinforcement- and graph-learning for

158 networked operational decisions, and the method foundations of continuous-
159 control reinforcement learning and graph neural networks, closing with the
160 design implications that motivate our approach. Section 4 presents the graph-
161 aware deep reinforcement learning framework. Section 5 describes the com-
162 putational experiments and results. Section 6 concludes the study and iden-
163 tifies future research directions.

164 **2. Distributed Network Manufacturing and Capacity Planning Prob-** 165 **lem**

166 In this section, we formulate the capacity planning problem for distributed
167 manufacturing systems as a production simulation model based on a Markov
168 decision process (MDP). Suppose the distributed manufacturing system con-
169 sists of N manufacturing facilities that produce one type of make-to-order
170 PRM product. Each manufacturing facility has its own production capacity,
171 including bioreactors and reagent inventory, and can purchase reagents from
172 source material suppliers. Facilities receive patient-indexed specimens and
173 initiate production when the matched specimen, an available bioreactor, and
174 sufficient reagents are co-located. Reagent transshipment and idle-bioreactor
175 relocation are allowed among qualified facilities. In addition, one intact wait-
176 ing specimen lot may be routed before manufacturing without changing the
177 patient-specimen association; pooling, splitting, substitution, cross-patient
178 use, and in-process transfer remain prohibited, and finished product is re-
179 turned to the same patient’s collection site (Section 2.3). Demand arrivals
180 and supplier disruptions are modeled as stochastic processes, implying that
181 local supply may be either insufficient or excessive relative to local demand

Table 1: Notation used in the distributed manufacturing and capacity planning model.

Sets and parameters	
Γ	Planning horizon indexed by t .
T	Manufacturing time for one PRM product, in periods, where $T < \Gamma$.
N	Set of manufacturing facilities indexed by n .
$\mathcal{E}_S$	Qualified facility pairs along which an intact waiting specimen lot may be routed before manufacturing.
m_t^n	Products that begin manufacturing at facility n at epoch t .
d_t^n	Demand arrivals at facility n in epoch t .
State variables	
s_t^n	Patient-indexed specimens waiting for manufacturing at their current facility n at epoch t .
r_t^n	Available reagent units at facility n at epoch t .
$\mathbf{b}_t^n$	Bioreactor work-in-process vector $(b_t^{n,1}, b_t^{n,2}, \dots, b_t^{n,T})$, where $b_t^{n,\tau}$ denotes the number of bioreactors at facility n with τ periods remaining at epoch t .
P_t^n	Maximum reagent purchase quantity for facility n for epoch $t + 1$.
Decision variables	
w_t^n	Executed integer net flow of intact waiting specimen lots into (positive) or out of (negative) facility n at epoch t , with identity and network conservation enforced (Section 2.3).
e_t^n	Reagent units added to or subtracted from facility n at epoch t and received or removed before epoch $t + 1$.
q_t^n	Idle bioreactors added to or subtracted from facility n at epoch t and received or removed before epoch $t + 1$.
p_t^n	Reagent purchase quantity for facility n for epoch $t + 1$.
Patient-condition and perishability	
H_j	Latent health index of patient j , drawn $H_j \sim \text{Uniform}(0, 1)$ at enrollment.
τ_j	Deterioration epoch of patient j , drawn $\tau_j \sim \theta \text{Weibull}(\kappa)$.
κ, θ	Shape and scale of the Weibull deterioration-epoch distribution.
α_j	Number of epochs since patient j enrolled, including waiting and manufacturing time.

182 at different decision epochs. The notation used throughout this section is
 183 summarized in Table 1.

184 Figure 1 illustrates the basic structure of the distributed manufacturing
 185 problem using a two-facility example. Each facility maintains a patient–
 186 specimen queue whose members carry a deteriorating survival index, together
 187 with reagent inventory, bioreactor capacity, and perishable finished product.
 188 At each decision epoch, condition-driven patient demand and disruption-
 189 prone reagent replenishment arrive under uncertainty. When one facility
 190 experiences excess demand, insufficient reagents, or limited production ca-
 191 pacity, the network may transship reagents, relocate idle bioreactors, or route
 192 an intact waiting specimen to a qualified manufacturing facility. The latter
 193 is patient-indexed logistics rather than sharing: identity, ownership, and the
 194 final infusion destination never change. This structure motivates the MDP
 195 formulation, where the state captures facility-level demand, patient condi-
 196 tion, material location and transit, inventory, and capacity, and the action
 197 coordinates both patient-indexed routing and fungible-resource decisions.

198 For the N facilities, T epochs of manufacturing time are required to
 199 produce one of the PRM products, and the planning horizon consists of Γ
 200 epochs where $\Gamma > T$.

201 At epoch $t \in \{1, 2, \dots, \Gamma\}$, the production status of the n -th facility for
 202 $n = 1, \dots, N$ is expressed as a state vector $\mathbf{x}_t^n$, where

$$\mathbf{x}_t^n = (s_t^n, r_t^n, \mathbf{b}_t^n, P_t^n), \quad (1)$$

203 Based on the state variables at epoch t , the facilities need to take action
 204 to adjust their production capacity for epoch $t + 1$. We define the action

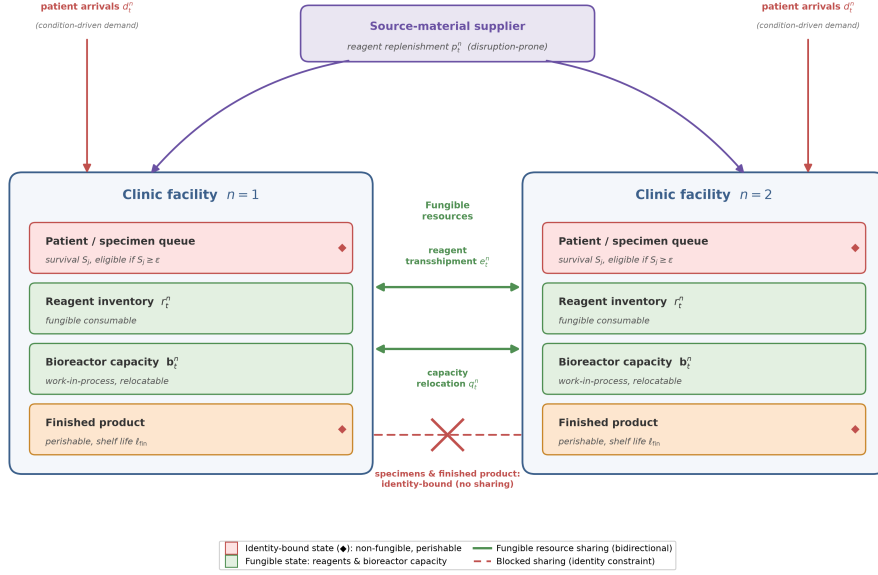


Figure 1: Illustration of a two-facility distributed manufacturing system for personalized regenerative medicine. Each facility holds a patient–specimen queue whose members carry a deteriorating survival index S_j (eligible while $S_j \geq \epsilon$), reagent inventory r_t^n , bioreactor capacity b_t^n , and perishable finished product. Condition-driven patient demand d_t^n and disruption-prone reagent replenishment p_t^n arrive under uncertainty. Reagents and idle bioreactor capacity are fungible and can be rebalanced through transshipment e_t^n and relocation q_t^n (solid green links). The crossed red link denotes prohibited pooling, substitution, or cross-patient reassignment of specimens and products; it does not prohibit identity-preserving transport of one intact waiting specimen lot to a qualified manufacturing facility or return of its product to the same patient.

205 $\mathbf{a}_t^n$ as a vector that consists of the amount of capacity sharing and reagent
 206 replenishment. The action is decided at the beginning of the production
 207 period t and executed before the start of $t + 1$.

$$\mathbf{a}_t^n = (w_t^n, c_t^n, q_t^n, p_t^n), \quad (2)$$

208 The goal of capacity planning is to minimize total production cost by
 209 dynamically selecting the actions from Equation (2) for all facilities, denoted
 210 by $\mathbf{A}_t = (\mathbf{a}_t^1, \dots, \mathbf{a}_t^N)$, for each epoch $t \in \Gamma$. We define $V(\mathbf{X}_t)$ as the total
 211 future cost of the selecting actions $\mathbf{A}_t$ when the states from all facilities,
 212 denoted by $\mathbf{X}_t = (\mathbf{x}_t^1, \dots, \mathbf{x}_t^N)$, are observed at epoch $t \in \Gamma$. The objective
 213 function can be expressed as a dynamic programming problem:

$$V(\mathbf{X}_t) = \min_{\mathbf{A}_t} E[c(\mathbf{A}_t; \mathbf{X}_t) + \gamma V(\mathbf{X}_{t+1})], \quad (3)$$

214 which is useful for finding the optimal action $\mathbf{A}_t$ for each state $\mathbf{X}_t$ at
 215 each epoch $t \in \Gamma$, where $c(\mathbf{A}_t; \mathbf{X}_t)$ is the immediate cost at time t , and
 216 $\gamma \in [0, 1]$ is a discount factor for balancing the impact of future costs in
 217 deciding actions. The choice of γ depends on the user's goal; if γ is closer
 218 to 1 (to 0), then it means the user cares more about long (short) term
 219 cost; if $\gamma = 0$, then only the cost for current epoch is considered. Equation
 220 (3) generalizes the optimality equation for the capacity planning problem in
 221 Li(2021) [6] in that cost functions $\mathbf{c}$ and $\mathbf{V}$ and the cumulative distribution
 222 function for the expectation operator can be unknown.

223 Equations (1)–(3) describe the base operational MDP shared with generic
 224 networked replenishment problems. The remainder of this section extends it
 225 along the three dimensions that distinguish PRM capacity planning, namely

the deteriorating clinical condition of the patients awaiting treatment, the hard expiry of biological source material and finished product, and the autologous identity constraint that forbids pooling of patient-specific specimens. Section 4 then develops a graph-aware DRL approach that learns the cost function $\mathbf{c}$, the value function $\mathbf{V}$, and the expectation operator directly from simulated experience.

2.1. Patient condition and treatment eligibility

In generic capacity planning, a unit of unmet demand is deferred, back-ordered, or lost at a fixed penalty, and its value does not change while it waits. PRM demand behaves differently, because each arrival is a patient whose clinical condition deteriorates throughout the vein-to-vein pathway. A specimen that waits or travels too long, or a therapy whose matched patient deteriorates during manufacturing or product return, may no longer produce a deliverable treatment. We therefore attach an explicit condition process to every enrolled patient from queue entry through specimen transit, manufacturing, and product return, adapting the real-time patient-health modeling of Tseng et al. [24] to a distributed capacity-planning setting.

When a patient j enrolls at facility n , we draw a latent health index $H_j \sim \text{Uniform}(0, 1)$ and a deterioration epoch $\tau_j \sim \theta \text{Weibull}(\kappa)$ with shape κ and scale θ ; H_j captures the heterogeneous baseline robustness across patients, while τ_j marks the onset of accelerated decline. Let α_j be the number of epochs since enrollment, including both queueing and manufacturing time. The patient’s survival index $S_j \in [0, 1]$ (a normalized proxy for the likelihood that the patient remains treatable) evolves as

$$S_j(\alpha_j) = \ell(\alpha_j) + H_j(u(\alpha_j) - \ell(\alpha_j)), \quad (4)$$

250 where $u(\alpha) = \exp(-\lambda_H g(\alpha))$ and $\ell(\alpha) = \exp(-\lambda_F g(\alpha))$ are the survival
 251 trajectories of an ideally healthy and a maximally frail patient, with decay
 252 rates $\lambda_H < \lambda_F$, so that H_j interpolates each patient between the two bounds.
 253 The effective exposure time

$$g(\alpha) = \begin{cases} \alpha, & \alpha \leq \tau_j, \\ \tau_j + \rho(\alpha - \tau_j), & \alpha > \tau_j, \end{cases} \quad (5)$$

254 accelerates by a factor $\rho > 1$ once the patient passes the deterioration epoch
 255 τ_j , reflecting the clinically observed transition to rapid decline. A patient
 256 remains eligible for treatment only while $S_j \geq \epsilon$. Before production starts,
 257 crossing the threshold removes the patient and specimen from either the
 258 waiting or transit state. After production starts, the same condition process
 259 continues through manufacturing and any modeled product-return epochs.
 260 If the matched patient becomes ineligible before completion, the in-process
 261 therapy is discarded and the occupied bioreactor is cleaned during that epoch
 262 before returning to the idle pool at the next decision epoch. Treatment ser-
 263 vice is recorded only when the finished product reaches and is assigned to the
 264 original patient, not when production starts. Because condition erodes mono-
 265 tonically in expectation and accelerates after τ_j , waiting, specimen transit,
 266 manufacturing, and product-return delay are not neutral buffering decisions.
 267 Parameter values are reported in Section 5.

268 2.2. Perishability of materials and finished product

269 Both the biological source material and the finished product are per-
 270 ishable and identity-specific, so they cannot be stockpiled against future de-
 271 mand. A waiting or in-transit specimen that is not put into production within
 272 ℓ_{mat} epochs of enrollment expires, so the patient is lost and the reserved ma-
 273 terial is wasted. Finished product that is not delivered within ℓ_{fin} epochs of
 274 completion likewise expires and is discarded. We track, per facility n and
 275 epoch t , patients lost to waiting-stage ineligibility $\phi_t^{n,\text{wait}}$, manufacturing-
 276 stage ineligibility $\phi_t^{n,\text{mfg}}$, return-stage ineligibility $\phi_t^{n,\text{ret}}$, local specimen ex-
 277 piry $\phi_t^{n,\text{exp}}$, transit ineligibility $\phi_t^{n,\text{tr-inel}}$, transit expiry $\phi_t^{n,\text{tr-exp}}$, and modeled
 278 transit loss $\phi_t^{n,\text{tr-loss}}$, together with finished units lost to expiry $\psi_t^{n,\text{fin}}$. A
 279 manufacturing-stage ineligibility also creates one discarded in-process ther-
 280 apy. It is convenient to collect the total patients lost and total material
 281 wasted at each facility,

$$\begin{aligned}
 \phi_t^n &= \phi_t^{n,\text{wait}} + \phi_t^{n,\text{mfg}} + \phi_t^{n,\text{ret}} + \phi_t^{n,\text{exp}} \\
 &\quad + \phi_t^{n,\text{tr-inel}} + \phi_t^{n,\text{tr-exp}} + \phi_t^{n,\text{tr-loss}}, \\
 \psi_t^n &= \phi_t^{n,\text{exp}} + \phi_t^{n,\text{mfg}} + \phi_t^{n,\text{tr-exp}} \\
 &\quad + \phi_t^{n,\text{tr-loss}} + \psi_t^{n,\text{fin}},
 \end{aligned} \tag{6}$$

282 where an expired or lost specimen contributes both a lost patient and a unit of
 283 wasted material. To signal impending losses before they occur, we also count
 284 the *at-risk unserved* patients whose survival has fallen within an urgency
 285 margin δ of the eligibility boundary, $v_t^n = |\{j \text{ waiting at } n : S_j < \epsilon + \delta\}|$.

286 2.3. Autologous identity constraint and admissible transfers

287 The autologous nature of PRM imposes a constraint with no analog in
 288 commodity supply chains. A specimen and the product manufactured from it
 289 are bound to one patient and cannot be substituted, pooled, split, or delivered
 290 to anyone else. Identity preservation does not, however, require the source
 291 material to remain at its collection site. We therefore allow *patient-indexed*,
 292 *identity-preserving*, *pre-manufacturing specimen routing*: only a complete
 293 lot in the waiting state may move, only along a qualified edge in $\mathcal{E}_S$, and at
 294 most once by a direct route before manufacturing begins. Multi-hop rout-
 295 ing, cycles, in-production transfer, cross-patient consumption, and product
 296 reassignment are prohibited.

297 The aggregate decision w_t^n records the executed integer net flow produced
 298 by a patient-level routing executor. The executor pairs facilities with oppo-
 299 site signed requests, selects eligible waiting identities deterministically, and
 300 conserves lots so that $\sum_n w_t^n = 0$. Each selected patient’s immutable identity
 301 record retains the collection facility, current material location, manufacturing
 302 facility, and transfer history. The main routing setting dispatches at epoch
 303 t , advances condition and specimen age once during one transit epoch, and
 304 makes the lot available before the epoch $t + 1$ production decision. After
 305 off-site manufacturing, the finished product remains assigned to the same
 306 patient and is returned to that patient’s collection or infusion facility. Thus,
 307 routing changes manufacturing location without creating unrestricted speci-
 308 men sharing.

309 2.4. Extended cost and the constrained graph MDP

310 The immediate cost augments the operational base cost c_{op} (covering
311 reagent purchase, holding, and shortage; bioreactor holding and shortage;
312 and reagent and capacity transfer) with three condition- and perishability-
313 driven terms:

$$c(\mathbf{A}_t; \mathbf{X}_t) = c_{\text{op}}(\mathbf{A}_t; \mathbf{X}_t) + \omega_{\text{lost}} \sum_{n \in N} \phi_t^n + \omega_{\text{exp}} \sum_{n \in N} \psi_t^n + \omega_{\text{urg}} \sum_{n \in N} v_t^n, \quad (7)$$

314 with weights ordered $\omega_{\text{lost}} \gg \omega_{\text{exp}} > \omega_{\text{urg}}$ to encode the clinical priority, in
315 which losing a patient dominates wasting material, which in turn dominates
316 the shaping penalty for letting patients drift toward the eligibility boundary.
317 The urgency term v_t^n steers the controller to act pre-emptively (starting
318 production or relocating capacity) before losses are actually incurred.

319 For clarity, we distinguish three quantities throughout the paper. For a
320 policy π , the expected total weighted objective is

$$J_{\text{tot}}^\pi = \mathbb{E}_\pi \left[\sum_{t=0}^{\Gamma-1} c(\mathbf{A}_t; \mathbf{X}_t) \right] = J_{\text{op}}^\pi + J_{\text{pat}}^\pi, \quad (8)$$

321 where J_{op}^π accumulates reagent purchase, holding, and shortage; bioreactor
322 holding and shortage; and reagent and capacity transfer, whereas J_{pat}^π accu-
323 mulates the weighted patient-loss, expiry, and urgency terms in Equation (7).
324 The total objective is the primary optimization and evaluation criterion. The
325 two component groups are reported separately to show which decisions gen-
326 erate an improvement.

327 All three quantities are reported in *modeled weighted-objective units*. Resource-
328 price coefficients can be interpreted as monetary equivalents under the adopted

329 calibration, but the patient-loss, expiry, and urgency weights are penalty
 330 or shadow-value parameters rather than observed accounting expenditures.
 331 Consequently, an absolute difference in J_{tot} is not presented as realized cash
 332 savings without an external economic calibration. Some large components
 333 are driven partly by exogenous demand and patient deterioration and by
 334 structural constraints such as finite capacity and lead times. They are there-
 335 fore only partly controllable, not “fixed” or wholly unavoidable.

336 For an anchor policy π_{anc} and candidate π , the headline relative improve-
 337 ment retains the complete objective as its denominator,

$$R_{\text{tot}} = \frac{J_{\text{tot}}^{\pi_{\text{anc}}} - J_{\text{tot}}^{\pi}}{J_{\text{tot}}^{\pi_{\text{anc}}}}. \quad (9)$$

338 We accompany it with absolute paired differences, component-specific rela-
 339 tive changes, and clinical outcomes. When a valid oracle or lower-cost refer-
 340 ence π_{ref} is available, we additionally report the fraction of anchor headroom
 341 captured,

$$R_{\text{headroom}} = \frac{J_{\text{tot}}^{\pi_{\text{anc}}} - J_{\text{tot}}^{\pi}}{J_{\text{tot}}^{\pi_{\text{anc}}} - J_{\text{tot}}^{\pi_{\text{ref}}}}. \quad (10)$$

342 This decomposition prevents a large patient-penalty or capacity-shortage
 343 component from obscuring a meaningful change in a directly affected op-
 344 erating component, while avoiding a post hoc replacement of the primary
 345 denominator.

346 To act on this cost, the controller needs visibility into the condition of
 347 each waiting queue. We therefore extend the facility state of Equation (1)
 348 with a condition summary $\boldsymbol{\sigma}_t^n$,

$$\tilde{\mathbf{x}}_t^n = (s_t^n, r_t^n, \mathbf{b}_t^n, P_t^n, \boldsymbol{\sigma}_t^n), \quad (11)$$

where σ_t^n contains the waiting count, waiting mean survival, near-expiry count, waiting survival-band histogram, specimens in transit to the facility, their mean survival, transferred specimens waiting at the facility, route-eligible waiting specimens, in-production count, in-production mean survival, and in-production at-risk count. The controller can therefore anticipate queue, transit, and manufacturing risk rather than merely react to aggregate demand.

Finally, we cast the extended problem as a *constrained graph MDP*. The facilities and their admissible information, reagent, capacity, and specimen-logistics links form a graph $\mathcal{G} = (N, \mathcal{E})$, with the qualified specimen edge set $\mathcal{E}_S \subseteq \mathcal{E}$ enforced separately from fungible-resource feasibility. The global state $\tilde{\mathbf{X}}_t = (\tilde{\mathbf{x}}_t^1, \dots, \tilde{\mathbf{x}}_t^N)$ is a signal over the nodes of $\mathcal{G}$, and the action $\mathbf{A}_t$ assigns per-node purchases, fungible-resource transfers, and conserved patient-indexed specimen-routing requests. The optimality equation (3) carries over verbatim with the extended state $\tilde{\mathbf{X}}_t$ and cost (7), but the value function must now trade current operating and logistics cost against the future risk of waiting, transit, patient loss, and expiry. This constrained graph MDP (perishable, identity-bound, and patient-condition-driven) is the problem the remainder of the paper addresses, and its graph structure is precisely what the method of Section 4 is designed to exploit.

3. Literature Review

We review three bodies of work that our approach draws together, namely the operations of personalized regenerative medicine (PRM) and decentralized manufacturing; reinforcement learning and graph learning for networked

operational decisions; and the underlying method foundations of continuous-control reinforcement learning and graph neural networks. Rather than surveying each exhaustively, we read them as an argument in which each stream supplies part of what a distributed PRM capacity-planning controller needs, none supplies all of it, and together they imply a small set of design choices that we make explicit at the end of the section. Throughout, we treat the learning machinery as established and concede that graph representation and reinforcement learning have already been combined for supply-chain problems; what remains open is the PRM problem itself, one that is perishable, identity-bound, and patient-condition-driven.

3.1. *Personalized regenerative medicine and decentralized manufacturing*

PRM supply chains differ substantially from conventional make-to-stock manufacturing systems because products are individualized, time-sensitive, and tightly coupled with patient-specific clinical pathways. In autologous cell therapy, patient material must be collected, transported, manufactured, quality-tested, and returned for administration, and manufacturing can begin only when a specimen, consumables, and production capacity are jointly available. These characteristics make fulfillment time, service reliability, and resource utilization central operational measures. Wang et al. [1] used simulation to compare centralized and point-of-care autologous cell therapy supply chain strategies and evaluated cost, fulfillment time, service level, and resource utilization. Li and White [6] further studied decentralized autologous cell therapy manufacturing networks and showed that transshipment and relocatable capacity can reduce the cost of resilience under supplier disruption. Most directly relevant to our patient-condition modeling, Tseng et al. [24]

integrated real-time patient health data into an autologous cell-therapy simulation and showed that representing patient condition, rather than treating demand as static, changes fulfillment and eligibility outcomes. We adopt this patient-condition perspective as a first-class driver of demand and feasibility, but embed it within a networked capacity-planning formulation solved by graph-aware learned control, rather than evaluating fixed policies in a simulation study alone.

These studies establish that PRM network design and inter-facility coordination are first-order engineering decisions rather than implementation details. Related work on cell therapy manufacturing process control also highlights the potential of reinforcement learning in uncertain, nonlinear, and partially observed manufacturing environments [2]. Together, this literature motivates a data-driven decision-support framework that can represent both the temporal dynamics of patient-specific manufacturing and the spatial dependencies created by distributed facilities.

3.2. Reinforcement learning for supply chain and manufacturing decisions

Reinforcement learning has been widely studied for sequential supply chain and manufacturing decisions. Early work applied RL to inventory replenishment in multi-echelon supply chains and perishable inventory systems [25, 26]. More recent studies have used deep reinforcement learning to handle high-dimensional state spaces, stochastic demand, and simulation-based decision environments. For example, Oroojlooyjadid et al. [27] developed a deep Q-network approach for the beer game, demonstrating that DRL can learn replenishment policies in decentralized, partially observed supply chain settings.

Recent supply chain RL studies have moved toward more realistic constraints and engineering settings. Bermudez et al. [19] proposed distributional constrained reinforcement learning for multi-period supply chain optimization with production and inventory constraints. Eisenach et al. [20] studied shared-resource capacity control in inventory management using a neural coordination mechanism and a reinforcement learning buying policy. Stranieri et al. [21] benchmarked classical and deep reinforcement learning inventory policies for pharmaceutical supply chains with perishability and non-stationary demand. These studies support the use of simulation-trained RL policies for uncertain operational control, but most focus on replenishment or inventory management rather than the joint capacity, reagent, and replenishment decisions (under a perishable, identity-bound demand process) that arise in distributed PRM manufacturing.

3.3. *Structure-informed, residual, and baseline-relative reinforcement learning*

Inventory-control evidence does not support treating generic end-to-end DRL as automatically superior to established operating rules. Gijsbrechts et al. [28] found that carefully tuned DRL can match state-of-the-art policies across lost-sales, dual-sourcing, and multi-echelon problems, while the invited roadmap of Boute et al. [29] argues that learning methods should exploit the structural policy knowledge accumulated in inventory research. De Moor et al. [30] made this principle operational by using strong perishable-inventory heuristics as teacher policies for reward shaping, improving training speed, stability, and, in suitable instances, performance relative to the teacher. Hybrid decomposition offers another route: Stranieri et al. [31] assigned pro-

448 duction decisions to DRL and shipping decisions to stochastic programming,
 449 obtaining more reliable performance than pure DRL. Non-stationary-demand
 450 experiments further show the value of conditioning policies on forecast infor-
 451 mation rather than assuming a fixed arrival law [32].

452 Residual reinforcement learning provides the most direct methodologi-
 453 cal foundation for our controller. The residual formulation decomposes the
 454 executed action into a competent base-controller action and a learned cor-
 455 rection, reducing the part of the control law that must be discovered from
 456 experience [33]. Staessens et al. [34] showed in an industrial-control setting
 457 that bounding the correction relative to a conventional controller can pre-
 458 serve its stabilizing role while adapting to uncertain operating conditions.
 459 Baseline-relative policy-improvement methods likewise motivate accepting a
 460 learned update only when there is credible evidence of improvement over
 461 the incumbent policy [35]. In stochastic inventory control, Deep Controlled
 462 Learning combines approximate policy iteration with common random num-
 463 bers and statistically efficient candidate selection to overcome the difficulty
 464 generic DRL has in beating strong heuristics [36]. These ideas motivate our
 465 MDL-2 anchor, bounded graph residual, common-random-number advantage
 466 filtering, and held-out deployment calibration. Our validation fallback is an
 467 empirical deployment safeguard, however, and should not be interpreted as
 468 inheriting the formal monotonic-improvement or closed-loop stability guar-
 469 antees proved under the assumptions of those methods.

470 3.4. *Deep reinforcement learning for regenerative medicine capacity planning*

471 The closest antecedent to this work is Tseng et al. [7], who developed a
 472 deep reinforcement learning approach for dynamic capacity planning in de-

473 centralized regenerative medicine supply chains. Their study demonstrated
474 that simulation-trained policies can adapt to demand and supply uncertainty
475 without requiring exact prior knowledge of the underlying stochastic pro-
476 cesses. This is directly aligned with the objective of using data-driven policy
477 learning for PRM capacity planning.

478 The present study builds on this line of work by adding an explicit graph
479 representation layer to encode inter-facility relationships. In decentralized
480 PRM networks, facilities are linked through reagent transshipment, idle-
481 bioreactor relocation, and qualified patient-indexed specimen logistics, while
482 every specimen and finished product remains bound to one patient. There-
483 fore, a facility’s decision depends not only on its own local state but also
484 on the states, route-eligible queues, transit pipeline, and fungible-resource
485 conditions of connected facilities. A flat vectorized state representation can
486 include all facility variables, but it does not explicitly exploit the graph struc-
487 ture that governs how information, material, and resources propagate across
488 the network. This motivates integrating graph convolutional representation
489 learning with continuous-control actor–critic reinforcement learning.

490 *3.5. Multi-agent reinforcement learning and graph learning for networked* 491 *supply chains*

492 Recent multi-agent reinforcement learning studies are relevant because
493 distributed manufacturing networks naturally involve decentralized infor-
494 mation and coordinated actions. Mousa et al. [18] formulated decentral-
495 ized inventory control as a multi-agent problem and studied centralized-
496 training/decentralized-execution policies under local information constraints.
497 Leluc et al. [37] similarly demonstrated the promise of cooperative multi-

498 agent reinforcement learning for inventory management under stochastic de-
499 mand and lead times. These studies show that decentralization is not only
500 an application feature but also an algorithmic design dimension.

501 In parallel, graph neural networks have emerged as a natural repre-
502 sentation tool for supply chains and other networked operational systems.
503 Chang et al. [14] learned production functions on supply-chain graphs us-
504 ing temporal graph neural networks and an inventory module, showing the
505 value of explicit relational structure for supply-chain inference and fore-
506 casting. Ahn et al. [15] used attention-based graph neural networks for
507 probabilistic supply and inventory prediction in enterprise supply-chain net-
508 works, while Ahn et al. [16] combined graph neural networks, offline re-
509 inforcement learning, and policy simulation for dynamic supply planning.
510 Kotecha and del Rio Chanona [13] provide the closest graph-control an-
511 tecedent: their GNN-MARL framework dynamically parameterizes heuris-
512 tic inventory policies and uses graph structure to coordinate decentralized
513 supply-chain agents. AFR-GCN-DDPG differs by learning a bounded con-
514 tinuous correction around a fixed look-ahead anchor, filtering correction tar-
515 gets through common-random-number look-ahead, and controlling patient-
516 condition, capacity, fungible-resource, and patient-indexed specimen-routing
517 decisions in an identity-bound PRM network rather than inventory replen-
518 ishment alone. These studies collectively suggest that graph-based learning
519 is useful for networked supply chain systems; however, most existing work
520 remains focused on prediction, inventory-only control, or general supply-
521 chain planning rather than continuous-action control of PRM-specific capac-
522 ity, reagent, and specimen flows.

523 *3.6. Method foundations: continuous-control reinforcement learning and graph*
524 *neural networks*

525 The actor proposes real-valued pressures for capacity relocation, reagent
526 replenishment and transfer, and specimen routing; a deterministic executor
527 converts those pressures into feasible mixed continuous and integer decisions.
528 This points to continuous-control actor-critic methods coupled with explicit
529 projection. Deterministic policy gradients and DDPG [8] extended actor-
530 critic learning to continuous action spaces, but are known to suffer from
531 value overestimation and training instability. TD3 [9] mitigates overestima-
532 tion through clipped double- Q learning, delayed policy updates, and target
533 smoothing; SAC [10] adds entropy regularization for more robust exploration
534 and stability; and PPO [11] is a widely used on-policy alternative built on a
535 clipped surrogate objective. The practical implication is that DDPG, though
536 historically the default for continuous supply-chain control, is no longer an
537 uncontested choice. A defensible study should compare backbones rather
538 than commit to one, and should expect the more stable off-policy methods
539 to reduce the run-to-run variance that plagues DDPG.

540 Graph neural networks encode relational structure through neighborhood
541 aggregation. The graph convolutional network [12] and its successors (Graph-
542 SAGE [38] for inductive neighborhood sampling and graph attention net-
543 works [39] for learned edge weighting) provide a permutation-equivariant
544 inductive bias that a flat multilayer perceptron lacks. Coupling such en-
545 coders with reinforcement learning has become a recognized paradigm for
546 networked decision problems [17]. For a clinic network whose facilities in-
547 teract through patient-indexed specimen logistics, capacity relocation, and

consumable transshipment, this relational bias is a modeling choice of substance rather than convenience, because it allows a single policy to reason over the topology along which patients, shortages, slack, and disruptions propagate.

3.7. Evaluation considerations for deep reinforcement learning

Evaluation rigor is especially important for DRL-based engineering applications. Henderson et al. [22] showed that deep reinforcement learning results can be sensitive to random seeds, hyperparameters, and implementation details. Agarwal et al. [23] further argued that few-run comparisons can be statistically unreliable and recommended more robust aggregate metrics and interval estimates. These findings imply that a DRL-based supply chain study should report implementation details, training budgets, out-of-sample test procedures, and variability across replications or random seeds when possible. In the present study, we therefore emphasize common simulation settings across all benchmark policies, detailed implementation parameters for the graph-aware actor-critic agents, and Monte Carlo evaluation under multiple disruption and demand forecast scenarios, with interquartile-mean performance and confidence intervals reported across random seeds.

3.8. Research gap and design implications

Read together, these literatures imply five design choices for a distributed PRM capacity-planning controller. First, because the decisions are sequential and the feasible action is mixed (continuous resource quantities together with projected integer specimen lots), a continuous proto-action actor-critic must be coupled to explicit decoding and feasibility projection rather than treated

Table 2: Positioning of this study relative to closely related literature.

Literature stream	Representative studies	Main limitation relative to this work
PRM operations	Simulation and decentralized planning [1, 6, 7]	No learned graph policy combining patient condition, perishability, identity, and coordinated continuous control.
Supply-chain RL	Inventory and constrained control [27, 19, 21, 20]	Primarily inventory or replenishment, not coupled PRM capacity and fungible-resource sharing.
Decentralized MARL	Cooperative inventory control [18, 37]	Not tailored to identity-bound, patient-specific make-to-order manufacturing.
Supply-chain GNNs	Graph forecasting and graph-aware planning [14, 15, 16, 13]	Mostly predictive or inventory-focused rather than continuous PRM network control.
Structured residual RL	Heuristic-guided and baseline-relative learning [30, 34, 35, 36]	No graph-structured PRM application with patient-condition risk and identity constraints.

572 as unconstrained real-valued control. Second, because a facility’s value de-
 573 pends on its network position and resources propagate along inter-facility
 574 links, its state should be encoded with a graph rather than a flat vector.
 575 Third, because strong operational rules already solve much of the nominal
 576 problem, the learned policy should target bounded, evidence-supported cor-
 577 rections to a transparent anchor rather than relearn the complete policy from
 578 scratch. Fourth, DDPG provides a natural canonical formulation and con-
 579 tinuity with earlier PRM capacity-planning work, but its known instability
 580 requires controlled comparison with TD3 rather than an untested assump-
 581 tion that DDPG is always best. Fifth, because reinforcement learning results
 582 are sensitive to seeds, implementation details, and the distinction between
 583 pretraining and online updates, evaluation should report paired multi-seed in-
 584 tervals, patient-service outcomes, and final-versus-frozen attribution checks.

585 None of these streams, however, addresses the problem that defines our
 586 setting. PRM operations research motivates adaptive, networked capacity
 587 planning but does not learn relational policies. Supply-chain graph-and-
 588 reinforcement-learning research supplies the relational-control machinery but
 589 treats demand as fungible and abstract, targeting forecasting or inventory-
 590 only control. Cell-therapy reinforcement learning operates on a single biopro-
 591 cess rather than a clinic network. As a result, no prior work models the three
 592 defining properties of PRM capacity planning (a perishable product with a
 593 hard viability deadline, an identity-bound product that cannot be pooled
 594 or substituted across patients, and demand driven by patient deterioration
 595 while waiting) within a single graph-aware continuous-control formulation.
 596 Our contribution closes this gap. We cast the problem as a constrained

graph MDP, build a reusable simulation testbed that realizes it, and evaluate graph-aware DDPG against strong flat-state and heuristic baselines, with matched TD3 as a development backbone ablation. SAC and PPO remain implemented family extensions rather than routing-primary empirical claims.

4. Graph-Aware Deep Reinforcement Learning Framework

Overview: a shared graph encoder with interchangeable actor-critic backbones

Distributed manufacturing networks for personalized regenerative medicine (PRM) comprise complex, interconnected facilities whose operating states are naturally represented as graphs. A flat controller can observe all facility features, but it must learn topology from position-dependent concatenation and lacks the permutation-equivariant neighborhood aggregation supplied by a GCN. We therefore adopt a two-part design. A graph convolutional network encoder maps the networked state to neighborhood-aware node embeddings, and a residual actor-critic maps those embeddings to proto-actions that are decoded and projected into feasible specimen, capacity, reagent, and replenishment decisions. Writing the controller as *encoder* $\circ$ *backbone* lets matched experiments isolate graph-versus-flat representation and, separately, DDPG-versus-TD3 backbone effects while holding the anchor, action decoder, projection, training budget, and evaluation streams fixed.

Primary AFR-GCN-DDPG controller and backbone variants

The primary proposed controller is advantage-filtered residual GCN-DDPG, abbreviated AFR-GCN-DDPG. The acronym names the complete controller rather than a new actor-critic backbone: *residual* means that the DDPG

620 actor modifies, rather than replaces, the MDL-2 anchor; *advantage-filtered*
 621 means that nonzero teacher corrections are retained only when paired common-
 622 random-number look-ahead estimates a positive improvement over that an-
 623 chor. This filtering is a teacher-data construction and regularization step
 624 inside AFR-GCN-DDPG, not a separate deployed policy. The controller
 625 combines the relational inductive bias of a GCN with the deterministic
 626 continuous-control mechanism used in earlier PRM capacity-planning re-
 627 search, while reducing the burden of relearning a complete operational policy
 628 from scratch. Following residual and constrained-residual control [33, 34], the
 629 actor learns only a bounded correction to a transparent look-ahead anchor.
 630 The framework supports TD3, SAC, and PPO behind the same GCN encoder
 631 (Table 4), but the routing-primary evidence in this paper evaluates DDPG
 632 and one matched development-only TD3 ablation; SAC and PPO are not
 633 ranked. A GCN encodes relational structure but has no temporal memory of
 634 its own; the actor-critic component supplies the sequential decision dynam-
 635 ics, and the two together encode spatial correlations and temporal trade-offs
 636 jointly.

637 We therefore present the proposed AFR-GCN-DDPG controller in full
 638 (the anchor-plus-residual action, GCN encoder, actor-critic updates, advantage-
 639 filtered supervision, and feasibility projection) and then note where each
 640 stability-oriented variant modifies the critic target, policy parameterization,
 641 or update schedule; the graph encoder and feasibility-projection layer are
 642 identical across backbones.

643 Figure 2 shows the graph abstraction used by the proposed graph-aware
 644 framework. Each healthcare clinic is represented as a node with facility-

Table 3: Terminology for the proposed controller and its matched comparisons.

Term	Meaning and role in this study
GCN; DDPG; TD3; SAC; PPO	Established acronyms. Standard graph-encoding and actor-critic algorithms, used with their conventional meanings and cited source formulations.
MDL-2	Study baseline. Two-period mean-demand look-ahead heuristic and transparent anchor; the label is not a general reinforcement-learning acronym.
Advantage-filtered distillation	Training step. Common-random-number look-ahead retains a nonzero teacher correction only when it improves on the anchor. It is not a separate policy.
AFR	Study-specific prefix. Shorthand for the complete <i>advantage-filtered residual</i> design: anchor, bounded learned correction, teacher regularization, feasibility projection, and deployment calibration.
AFR-GCN-DDPG	Proposed controller. AFR with a GCN state encoder and DDPG actor-critic backbone.
AFR-Flat-DDPG; AFR-GCN-TD3	Matched comparisons. The first removes the graph encoder; the second changes the RL backbone. Neither is a separate proposed framework.

Table 4: Continuous-control actor-critic backbones evaluated behind the shared GCN encoder.

Backbone	Policy class	Sampling	Key mechanism and role in this study
DDPG [8]	Deterministic	Off-policy	Canonical backbone of the proposed AFR-GCN-DDPG controller; retains continuity with prior PRM capacity-planning work.
TD3 [9]	Deterministic	Off-policy	Stability-enhanced variant using clipped double- Q learning, delayed policy updates, and target smoothing.
SAC [10]	Stochastic	Off-policy	Entropy-regularized objective for more robust exploration and training stability.
PPO [11]	Stochastic	On-policy	Clipped surrogate objective; an on-policy alternative to the off-policy backbones.

645 specific state features, including waiting and inbound specimens and their
 646 condition summary, reagent inventory, available bioreactor capacity, and
 647 other local operational conditions. The edges encode different forms of inter-
 648 facility dependency. Information-flow edges let the model propagate oper-
 649 ational and patient-condition signals among neighboring clinics; resource-
 650 sharing edges represent feasible reagent transshipment relationships; capacity-
 651 sharing edges connect each clinic to the central capacity hub; and an explic-
 652 itly configured subset of qualified inter-clinic links permits identity-preserving
 653 routing of intact waiting specimen lots. By applying graph convolution over
 654 this structure, the model learns a network-aware representation in which each
 655 facility’s decision is informed not only by its own state but also by the states
 656 of connected facilities. This graph embedding is then passed to the actor-
 657 critic backbone to support coordinated specimen-routing, capacity, reagent,
 658 and replenishment decisions.

659 Formally, let $G_t = (V, E_t)$ represent the manufacturing network at epoch
 660 t , where each node $v_i \in V$ denotes a facility with state features x_i^t and edge
 661 weights e_{ij}^t indicate logistic or transshipment costs. The GCN encodes local
 662 and neighborhood information using a layer-wise propagation rule:

$$h_i^{(l+1)} = \sigma\left(W^{(l)} \sum_{j \in \mathcal{N}(i)} \frac{1}{c_{ij}} h_j^{(l)}\right), \quad (12)$$

663 where $h_i^{(0)} = x_i^t$, $W^{(l)}$ are trainable weights, $\sigma(\cdot)$ is a ReLU activation, and
 664 c_{ij} is a degree-based normalization factor. The resulting embedding $H_t =$
 665 $\{h_i^{(L)}\}_{i=1}^N$ captures the current global state of the network.

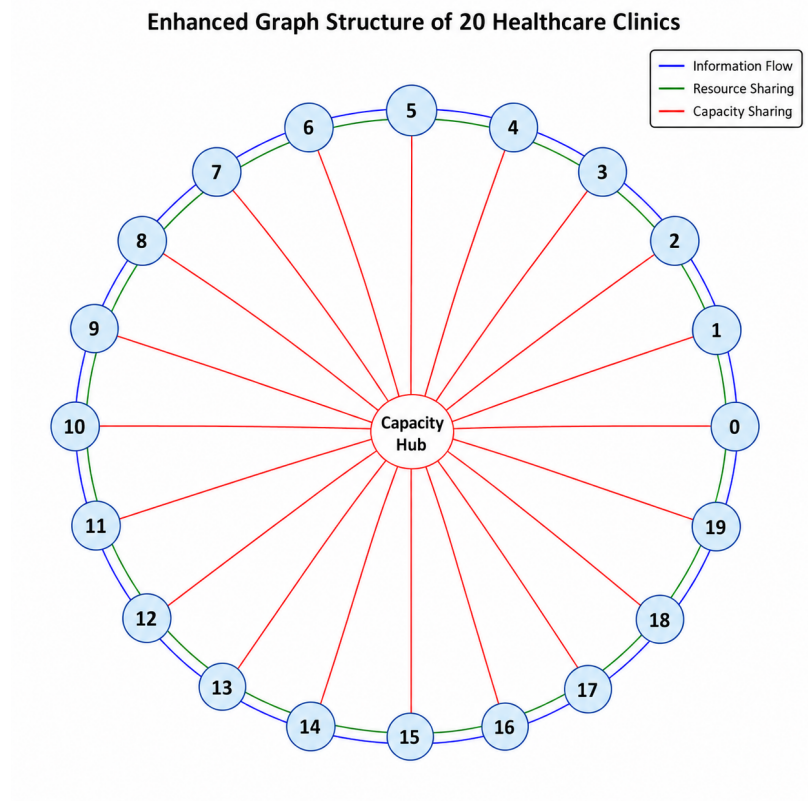


Figure 2: Graph representation of a distributed personalized regenerative medicine manufacturing network with 20 healthcare clinics. Each outer node represents a clinic or manufacturing facility, and the central node represents the shared capacity pool. Blue edges indicate information flow among neighboring clinics, green edges indicate fungible-resource relationships, and red radial edges indicate capacity sharing between each clinic and the central capacity hub. Qualified specimen-logistics links are configured as a constrained subset of inter-clinic relationships and are omitted from this conceptual edge legend. The graph convolutional network encodes local and neighboring state for downstream patient-indexed routing and resource decisions.

666 *Overview of the graph-aware actor-critic workflow (DDPG instantiation)*

667 Training has two explicit stages. First, common-random-number look-
 668 ahead constructs advantage-filtered residual targets relative to MDL-2 and
 669 initializes the graph encoder and actor on those targets. Second, DDPG in-
 670 teracts with the simulator, stores feasibility-projected actions and rewards in
 671 replay, and updates the critic, actor, and shared encoder while retaining the
 672 filtered teacher loss as an actor regularizer. At execution time, the observed
 673 node features change with queues, patient condition, inventories, disruptions,
 674 and transit pipelines; the configured clinic topology supplies the graph over
 675 which the same learned encoder aggregates those dynamic features. The
 676 GCN therefore processes a time-varying network state but does not infer a
 677 new physical network at every epoch.

678 This two-stage description is important for attribution. AFR-GCN-
 679 DDPG denotes the controller that results from the complete procedure,
 680 whereas final-versus-frozen comparisons separately test whether the online
 681 DDPG phase adds endpoint value beyond the advantage-filtered initializa-
 682 tion.

683 *Initialization*

- 684 • Critic network $Q(X_t, A_t|\theta_Q)$ initialized with weights θ_Q .
- 685 • Actor network $\mu(X_t|\theta_\mu)$ initialized with weights θ_μ .
- 686 • GCN initialized with weights ω for processing graph states.

687 *DDPG Component*

688 The DDPG agent operates on the GCN-encoded state H_t to generate
 689 a bounded residual action $\Delta A_t = \mu(H_t|\theta_\mu)$. Let A_t^{anc} denote the action

690 produced by the transparent look-ahead anchor and let ρ be a vector of
 691 decision-group-specific trust-region scales. The raw proposed action is

$$A_t^{\text{raw}} = A_t^{\text{anc}} + \rho \odot \Delta A_t. \quad (13)$$

This residual parameterization preserves the anchor’s basic inventory logic while allowing the GCN policy to adapt patient-indexed specimen routing, replenishment, and fungible resource-sharing decisions to network-wide patient pressure, geographic connectivity, disruptions, and demand-prior drift. The critic network $Q(H_t, A_t|\theta_Q)$ estimates expected discounted return, where reward is the negative modeled cost. The optimization follows the standard DDPG updates:

$$y_i = r_i + \gamma Q'(H_{i+1}, \mu'(H_{i+1}|\theta_{\mu'})|\theta_{Q'}), \quad (14)$$

$$L_Q = \frac{1}{N} \sum_i (y_i - Q(H_i, A_i|\theta_Q))^2, \quad (15)$$

$$\nabla_{\theta_\mu} J = \frac{1}{N} \sum_i \nabla_a Q(H_i, a|\theta_Q)|_{a=\mu(H_i)} \nabla_{\theta_\mu} \mu(H_i|\theta_\mu). \quad (16)$$

692 *Advantage-filtered residual initialization and deployment*

693 The useful residual corrections in AFR-GCN-DDPG are sparse because
 694 the MDL-2 anchor is already competitive in many states. We therefore ini-
 695 tialize the residual actor through advantage-filtered teacher construction. At
 696 a sampled state, we compare a small set of decision-group residual candi-
 697 dates, including specimen routing and network-resource actions, with the
 698 zero-residual MDL-2 action over a finite look-ahead horizon. Candidate and
 699 anchor rollouts use common random numbers so that their score difference is
 700 not dominated by different demand, deterioration, or disruption realizations.

701 A nonzero target is retained only when it improves the configured operating-
 702 cost and patient-outcome score; otherwise the target is the zero residual. To
 703 keep the sparse improvement labels from being overwhelmed, improving and
 704 zero-residual examples receive equal total weight. The weighted advantage-
 705 filtered data remain in the actor loss during DDPG fine-tuning rather than
 706 being used only as a one-time warm start.

707 Deployment uses two independent validation stages. First, checkpoints
 708 saved during training are compared on common validation scenarios. Second,
 709 the selected checkpoint is evaluated over a fixed set of residual trust-region
 710 scales, including zero. The nonzero residual is deployed only when it improves
 711 validation cost without reducing service; otherwise the controller falls back to
 712 MDL-2. This mechanism keeps the deployed controller within the residual
 713 GCN-DDPG family while preventing a poorly generalized correction from
 714 overriding the operational anchor.

715 *Reward function*

716 The agents optimize the negative of the immediate cost defined in Sec-
 717 tion 2.4. For each epoch t , the reward is

$$r_t = -c(\mathbf{A}_t; \tilde{\mathbf{X}}_t), \quad (17)$$

718 where $c(\cdot)$ is the extended cost of Equation (7), comprising the operational
 719 base cost c_{op} (reagent purchase, holding, and shortage; bioreactor holding and
 720 shortage; specimen and reagent logistics; and capacity relocation) together
 721 with the patient-loss, material-and-product-expiry, and urgency terms. Op-
 722 timizing this reward therefore requires the agent to trade current operating

723 and logistics cost against the future risk of patient losses and expiry, rather
724 than resource utilization alone.

725 *Action decoding and feasibility projection..* Because the actor network out-
726 puts continuous residuals, an additional deterministic decoding and feasibility-
727 projection step is used before actions are executed in the simulation environ-
728 ment. The raw anchor-plus-residual action A_t^{raw} is first scaled to the admis-
729 sible range of each decision variable and then clipped to its variable-specific
730 lower and upper bounds. The specimen block expresses signed facility pres-
731 sure rather than anonymous divisible flow. It is scaled by the maximum route
732 quantity and rounded half away from zero to integer requests; a shared ex-
733 ecutor then matches opposite requests only across qualified edges and selects
734 specific eligible patient identities. Reagent transshipment, bioreactor relo-
735 cation, and replenishment are decoded through their corresponding bounds
736 and feasibility rules.

737 The decoded action is subsequently passed through a repair operator that
738 enforces the operational constraints of the manufacturing network. First,
739 nonnegativity constraints are imposed so that the resulting action cannot
740 remove more reagents, idle bioreactors, or route-eligible waiting specimen
741 lots from a facility than are available at that epoch. Second, replenishment
742 quantities are capped by the maximum purchasable reagent quantity P_t^n and
743 by supplier availability. Third, network conservation is enforced separately
744 for fungible resources and integer specimen lots. A specimen route is legal
745 only for a waiting identity that has not previously moved; each executed
746 donor event has one matched receiver event, and unmatched requests are
747 recorded as blocked rather than repaired by substitution. Finally, all transfer

Algorithm 1 Graph-aware actor–critic for distributed PRM capacity planning (DDPG instantiation; TD3/SAC/PPO swap the critic target, policy parameterization, or update schedule).

Require: Anchor policy π_{anc} , residual candidates $\mathcal{D}$, look-ahead horizon L , discount factor γ , soft-update rate τ , actor and critic learning rates α_μ, α_Q , exploration scale σ , minibatch size N .

- 1: Initialize GCN encoder g_ω , actor $\mu(\cdot \mid \theta_\mu)$, critic $Q(\cdot, \cdot \mid \theta_Q)$, target networks μ' and Q' , and replay buffer $\mathcal{R}$.
 - 2: Build weighted teacher data by comparing $\pi_{\text{anc}}(X) + \delta$, $\delta \in \mathcal{D}$, with $\pi_{\text{anc}}(X)$ under common-random-number L -step rollouts; use target δ^* only when it improves the score and zero otherwise.
 - 3: Fit the residual actor to the weighted advantage-filtered targets and retain this loss as an actor regularizer.
 - 4: **for** each environment step t **do**
 - 5: Observe raw graph state X_t and compute graph embedding $Z_t = g_\omega(X_t)$.
 - 6: Select exploratory residual action $A_t = \pi_{\text{anc}}(X_t) + \rho \odot \mu(Z_t \mid \theta_\mu) + \epsilon_t$, where $\epsilon_t \sim \mathcal{N}(0, \sigma^2 I)$.
 - 7: Project or clip A_t to the feasible set and execute the resulting action $\hat{A}_t$.
 - 8: Observe cost/reward r_t , next state X_{t+1} , and store $(X_t, \hat{A}_t, r_t, X_{t+1})$ in $\mathcal{R}$.
 - 9: **if** an update is scheduled **then**
 - 10: Sample minibatch $(X_i, A_i, r_i, X_{i+1})_{i=1}^N$ from $\mathcal{R}$ and compute $Z_i = g_\omega(X_i)$, $Z_{i+1} = g_\omega(X_{i+1})$.
 - 11: Compute targets $y_i = r_i + \gamma Q'(Z_{i+1}, \mu'(Z_{i+1} \mid \theta_{\mu'}) \mid \theta_{Q'})$.
 - 12: Update critic by minimizing $L_Q = N^{-1} \sum_i (y_i - Q(Z_i, A_i \mid \theta_Q))^2$.
 - 13: Update actor using the deterministic policy gradient plus the weighted advantage-filtered regularizer.
 - 14: Backpropagate through the GCN encoder and softly update target networks.
 - 15: **end if**
 - 16: **end for**
 - 17: Select checkpoint and residual scale on independent validation scenarios; return the calibrated residual policy or the MDL-2 fallback.
-

748 and routing actions are restricted to their configured graph edges. Finished-
 749 product return is a deterministic identity-preserving lifecycle transition, not
 750 an actor decision.

751 Equivalently, the executed action can be interpreted as the feasible action
 752 $\hat{A}_t$ obtained by projecting the decoded actor output onto the feasible action
 753 set $\mathcal{F}(X_t)$:

$$\hat{A}_t = \Pi_{\mathcal{F}(X_t)}(A_t^{\text{raw}}), \quad (18)$$

754 where $\mathcal{F}(X_t)$ is determined by current facility states, patient lifecycle and
 755 route history, inventory availability, capacity availability, supplier limits, the
 756 autologous identity constraint, and graph connectivity. This projection step
 757 improves engineering credibility by ensuring that all actions executed in the
 758 simulation correspond to implementable specimen-routing, capacity, replen-
 759 ishment, and transshipment decisions.

760 Each stability-oriented backbone reuses this encoder and projection layer
 761 unchanged and alters only the reinforcement-learning update. TD3 replaces
 762 the single critic with clipped double- Q targets, delays actor updates, and
 763 adds target-policy smoothing; SAC replaces the deterministic actor with an
 764 entropy-regularized stochastic policy and a soft value target; and PPO re-
 765 places off-policy replay with on-policy rollouts optimized under a clipped
 766 surrogate objective. In all cases the actor and critic operate on the GCN
 767 embedding Z_t rather than on the flat state, and the executed action is the
 768 feasibility-projected $\hat{A}_t$.

769 *Summary*

770 The proposed method is therefore a residual GCN-DDPG controller: a
 771 GCN encoder supplies a permutation-equivariant, network-aware state repre-

772 sentation; a deterministic actor learns bounded corrections to a transparent
 773 look-ahead anchor; and a feasibility-projection layer guarantees that every
 774 executed action respects inventory, capacity, supplier, patient identity, route
 775 history, and connectivity constraints. The routing-primary architecture com-
 776 parison holds specimen-routing availability, the anchor, residual action space,
 777 advantage-filtered teacher protocol, projection, training budget, and evalu-
 778 ation seeds fixed between GCN and parameter-matched flat encoders. The
 779 earlier backbone comparison instead holds the graph encoder fixed while
 780 replacing DDPG with TD3. Residual SAC and PPO are secondary fam-
 781 ily extensions because their stochastic residual distributions and likelihood
 782 calculations require additional matched implementation work. These vari-
 783 ants test backbone robustness without redefining them as separate proposed
 784 methods.

785 **5. Computational Experiments**

786 This section reports three empirical stages designed to separate anchor
 787 improvement, graph representation, and online reinforcement-learning attri-
 788 bution. The first stage calibrates patient-condition dynamics and evaluates
 789 replenishment residuals under demand-prior drift. The second expands the
 790 action space to geographically structured reagent and idle-capacity transfers
 791 and compares the graph policy with a parameter-matched flat policy. The
 792 third holds the pretrained graph controller fixed while testing whether conser-
 793 vative online TD3 updates add value beyond frozen pretraining. The exper-
 794 iments compare the learned graph-aware correction policy with its heuristic
 795 anchor, other heuristic and look-ahead policies, flat-state residual policies,

796 and graph-aware backbone variants under the same simulation environment.

797 5.1. Objective

798 The objective of this study is to compare the proposed residual GCN–
799 DDPG framework with traditional heuristic methods, its MDL-2 anchor, flat-
800 state residual DDPG, and stability-oriented graph-aware actor–critic variants
801 in distributed manufacturing.

802 5.2. Methodology

803 5.2.1. Benchmarking Setup

804 The final benchmark is organized as a hierarchy of matched comparisons
805 rather than a single undifferentiated algorithm ranking (Table 5). The first
806 comparison tests whether reinforcement learning improves the MDL-2 an-
807 chor. The second isolates the graph representation by replacing the GCN
808 with a flat MLP while keeping the entire residual-learning recipe fixed. The
809 third tests whether the result is specific to DDPG by repeating the residual
810 comparison with TD3. Online rollout shields and the supervised GCN shield
811 selector are reported separately because the former uses additional simula-
812 tion at decision time and the latter is not an RL policy. Pure end-to-end RL
813 and the stochastic SAC/PPO family are reported as secondary comparators
814 rather than being allowed to substitute for these mandatory tests.

815 The heuristic baselines are:

- 816 • **MYO (Myopic):** This heuristic minimizes single-period cost without
817 considering future demand.

- 818 • **MDL-1 (One-period Mean Demand Lookahead):** MDL-1 ex-
819 tends the planning horizon to one lookahead period using deterministic
820 state transitions based on mean demand.
- 821 • **MDL-2 (Two-period Mean Demand Lookahead):** MDL-2 ex-
822 tends the planning horizon to two lookahead periods using determinis-
823 tic state transitions based on mean demand.
- 824 • **ISO (Isolated Regional Facilities):** ISO assumes no resource shar-
825 ing between facilities and focuses only on local replenishment decisions.
- 826 • **F-MYO (Forecast Myopic):** F-MYO augments current-period net-
827 work balancing with the demand forecast available at the decision
828 epoch.
- 829 • **uMYO and pMYO (Patient-aware myopic policies):** uMYO
830 adds urgency-based replenishment and capacity surges, while pMYO
831 uses a bounded patient-risk workload uplift. These controls prevent
832 the proposed method from being compared only with condition-blind
833 heuristics.

834 The learning-based comparisons are:

- 835 • **MLP residual DDPG / flat-state residual DDPG:** This base-
836 line uses the same MDL-2 anchor, residual action space, DDPG actor-
837 critic structure, training budget, and feasibility projection as the pro-
838 posed method, but removes the GCN encoder. Instead, all facility-
839 level states are concatenated into a flat vector and passed directly into

840 multilayer perceptron actor and critic networks. This baseline is de-
841 signed to answer whether performance improvements come from rein-
842 forcement learning alone or from explicitly encoding graph-structured
843 inter-facility dependencies.

844 • **Residual GCN–TD3 and matched flat residual TD3:** These poli-
845 cies keep the anchor, advantage-filtered supervision, action constraints,
846 training data, and evaluation streams fixed while replacing the DDPG
847 update with TD3. They test whether twin critics, delayed actor up-
848 dates, and target smoothing improve stability.

849 • **Pure GCN and flat actor–critic policies:** From-scratch DDPG
850 and TD3 quantify the benefit of anchoring; SAC and PPO provide
851 secondary off-policy stochastic and on-policy comparisons. Residual
852 SAC/PPO are included only after their residual-space probability cal-
853 culations and matched flat variants pass the same validation protocol.

854 In addition, graph ablation experiments are planned to evaluate the role
855 of different edge types. Specifically, we consider removing capacity-sharing
856 edges and removing resource-sharing edges to assess whether these relational
857 structures contribute to policy performance under disruption and demand
858 uncertainty.

859 All methods receive the same decision-time observations and are eval-
860 uated on identical paired random-number streams. In demand-prior-drift
861 experiments, heuristics receive the same misspecified demand prior available
862 to the controller and no method receives the realized future demand rates.
863 Learned policies use the same training-scenario distribution, environment-

Table 5: Final comparison hierarchy for the proposed controller.

Comparison layer	Required policies	Question answered
Anchor improvement	im- AFR-GCN-DDPG versus MDL-2	Does the learned correction improve the strong policy from which it starts?
Strong heuristics	heuristic- ISO, MYO, F-MYO, uMYO, pMYO, MDL-1, and MDL-2	Is the controller competitive with sharing, forecast-aware, patient-aware, and look-ahead rules?
Representation ablation	AFR-GCN-DDPG versus matched AFR-flat-DDPG	Is any gain attributable to graph representation rather than residual learning alone?
Backbone robustness	ro- Residual GCN-DDPG versus residual GCN-TD3, each with a matched flat counterpart	Does the result persist when the deterministic actor-critic backbone changes?
Anchor ablation	abla- Residual versus pure GCN/flat DDPG and TD3	Does the heuristic anchor improve sample efficiency, stability, and final performance?
Deployment reference	MDL-2/pMYO rollout shields and supervised GCN shield selector	How does policy quality trade off against online simulation cost, and can RL outperform non-RL distillation?
Secondary family	GCN/flat SAC and PPO; residual forms after matched implementation	Is the conclusion robust to stochastic off-policy and on-policy learning?
Graph mechanism	Full graph, no/shuffled edges and geographic-edge ablations	Does the policy use topology rather than merely additional parameters?

864 interaction budget, checkpoint-selection budget, and held-out evaluation seeds
 865 within each matched comparison. Online decision time and the number of
 866 simulator rollouts are reported separately for rollout-based methods. This
 867 prevents a performance difference from being created by unequal forecasts,
 868 simulator access, evaluation noise, or hidden online computation.

869 5.2.2. *Performance Metrics*

870 We evaluate each method using both objective-based and engineering-
 871 relevant performance metrics. The primary metric is the average total weighted
 872 objective J_{tot} in Equation (8). It includes the complete operating-cost group
 873 (reagent purchase, holding, and shortage; bioreactor holding and shortage;
 874 and reagent and capacity transfer) and the patient-related penalty group
 875 (patient loss, material or therapy expiry, and urgency). The percentage im-
 876 provement reported in headline comparisons always uses the anchor’s com-
 877 plete J_{tot} as the denominator, as in Equation (9); we do not remove large
 878 components after observing the result.

879 Because a weighted total can be dominated by patient-loss or capacity-
 880 shortage penalties, every formal comparison also reports the absolute paired
 881 difference and an additive cost decomposition. Component-specific percent-
 882 ages are secondary mechanism measures: they reveal whether a policy saves
 883 purchase, holding, shortage, transfer, or patient-penalty cost, but do not
 884 replace the primary total-objective comparison. When an oracle or other
 885 defensible lower-cost reference is evaluated, we also report the headroom-
 886 capture measure in Equation (10). To strengthen the application relevance
 887 of the experiments, we track the following secondary metrics:

- 888 • **Cost decomposition:** operating/base cost and its reagent, bioreac-

889 tor, and transfer components, together with patient-loss, expiry, and
890 urgency penalties.

- 891 • **Completion service level:** the fraction of eligible patient demand
892 completed within the planning horizon.
- 893 • **Patient loss and manufacturing-period ineligibility:** the number
894 of patients who become ineligible or expire before treatment and the
895 fraction who become ineligible after manufacturing begins.
- 896 • **Average waiting time / fulfillment delay:** the average number of
897 epochs that patient specimens wait before manufacturing begins.
- 898 • **Reagent shortage frequency:** the frequency with which manufac-
899 turing is delayed because insufficient reagents are available.
- 900 • **Bioreactor utilization:** the fraction of available bioreactor capacity
901 used for production across facilities.
- 902 • **Routing, transshipment, and relocation activity:** patient-indexed
903 specimen route count, distance, transit time, route cost, and blocked
904 requests, together with reagent movements and idle-bioreactor relo-
905 cations. These are mechanism diagnostics rather than evidence that
906 routing is itself a contribution.
- 907 • **Inference time:** the computational time required for a trained policy
908 or heuristic to generate one epoch-level decision.

909 These metrics are included because the objective of PRM capacity plan-
910 ning is not only to reduce a modeled weighted objective, but also to support

timely patient-specific manufacturing, effective resource utilization, and reliable system performance under demand and supply uncertainty. Cost-weight sensitivity is therefore required before translating an objective difference into an economic claim.

5.2.3. *Heuristic Implementations*

We provide brief descriptions of the heuristic methods implemented:

- **MYO (Myopic):** MYO focuses on minimizing costs within the current period without considering future consequences.
- **MDL-1 and MDL-2:** These methods extend the planning horizon to one and two lookahead periods, respectively, using deterministic state transitions based on mean demand.
- **ISO (Isolated Regional Facilities):** ISO assumes no sharing of resources between manufacturing facilities and primarily concentrates on reagent replenishment.

5.2.4. *Implementation and statistical protocol*

The models are implemented in Python using PyTorch, and all experiment parameters, training seeds, checkpoint candidates, deployment scales, and evaluation streams are stored in configuration files. Within a matched comparison, the graph and flat policies receive the same state variables, anchor actions, teacher targets, environment interactions, validation budget, and common-random-number (CRN) holdout episodes. The graph policy replaces the flat encoder with message passing over the clinic graph; the re-

933 maining actor–critic and deployment components are held fixed or parameter
934 matched.

935 *Routing-primary hyperparameter development.* The routing-primary DDPG
936 settings were obtained through sequential, development-only sensitivity screens
937 rather than an exhaustive automated HPO search. The screens first bounded
938 the residual correction and then varied the online update schedule, actor
939 learning rate, exploration noise, imitation and frozen-reference regularization,
940 and deployment gate. Mechanism-development screens separately examined
941 integer-lot straight-through specimen actions, anchor-relative rewards, re-
942 turn horizon, and evidence required to release the frozen-pretraining refer-
943 ence. Table 6 summarizes the principal ranges and the current candidate.
944 All screens used development seeds and CRN streams; none is included in a
945 confirmatory estimate.

946 The selected candidate uses batch size 64, critic learning rate 3×10^{-4} ,
947 integer specimen-lot execution with a straight-through actor gradient, and
948 specimen-only residual corrections; reagent, capacity, and replenishment resid-
949 ual groups remain zero in this screen. The most informative warning against
950 single-seed selection came from the update-frequency study: the frequency-4
951 candidate passed its seed-0 screen but did not generalize in the subsequent
952 three-seed confirmation. Seed 0 and the earlier routing seeds 1–2 are there-
953 fore treated as development evidence, not untouched confirmation seeds.

954 The replenishment-residual DDPG and TD3 confirmation uses 300 train-
955 ing episodes for each of five seeds, checkpoint candidates between episodes
956 50 and 300, and 100 independent paired holdout replications per seed. Re-
957 ported confidence intervals use a two-level bootstrap over training seeds and

Table 6: Routing-primary DDPG development screens. The selected values define the current candidate but are not themselves confirmatory results.

Component	Development values or variants	Selected value
Residual deployment magnitude	Anchor-only; 0.10, 0.25, and 0.50	0.10 for specimen transfer
Environment update frequency	Every 1 or 4 simulator steps	1
Actor update frequency	Every 2, 4, or 8 critic updates	2
Online actor learning rate	1×10^{-5} or 3×10^{-5}	1×10^{-5}
OU exploration standard deviation	0.005, 0.02, or 0.05	0.005
Imitation/reference regularization	Imitation 1 or 2; reference 0, 250, or 500	1 and 500, respectively
Pretraining/critic warm start	Short smoke or 300 epochs; 0 or 500 critic updates	300 epochs and 500 updates
Online return evidence	Immediate or four-step anchor-relative return; unfiltered, positive, or dual confirmation	Four-step dual confirmation
Material improvement threshold	Zero or one patient-loss-equivalent scaled cost	0.0005
Deployment gate	Fixed development validation grid with anchor fallback	0.575

958 paired evaluation replications. The regional network-action experiment uses
959 three independently initialized graph and flat policies, 30 paired validation
960 episodes, and 100 previously unseen paired holdout replications per seed.
961 Because its selected checkpoint is dominated by advantage-filtered teacher
962 distillation, that experiment is interpreted as graph-policy evidence rather
963 than as independent evidence for online DDPG updates.

964 The conservative online-TD3 screen starts from the corresponding frozen
965 pretrained graph and flat checkpoints and runs 100 online episodes for seeds
966 0, 1, and 2. Each run completed 5,200 update calls, including 5,073 critic
967 updates and 2,286 delayed actor updates. Final and frozen checkpoints are
968 evaluated with identical CRN keys, using 20 validation and 100 holdout
969 replications per seed for the final policies and 5 validation and the same 100
970 holdout replications per seed for the frozen controls. This campaign was run
971 on an NVIDIA RTX 4090 GPU; all optimization diagnostics were finite, and
972 no CPU fallback occurred. These experiment-specific budgets replace the
973 earlier generic one-million-step description and make the reported results
974 traceable to the executed configurations.

975 The routing-primary campaign is prospectively separated from those ear-
976 lier no-routing results. It holds patient-indexed specimen routing enabled
977 for every primary policy and compares GCN residual DDPG, a parameter-
978 matched flat residual DDPG controller, and the MDL-2 routing anchor.
979 One fresh routing-enabled teacher is shared across the learned architectures.
980 After the development screens are frozen, confirmatory training uses five
981 untouched seeds (10–14) for each learned architecture. Every run first re-
982 ceives 100 episodes, or approximately 5,200 online transitions, with complete

state saved every five episodes. Prespecified development learning curves at episodes 25, 50, 75, and 100 determine one global budget decision before the final holdout is opened: if either architecture has not plateaued, all ten runs are extended to 200 episodes. Extension to the common maximum of 300 is allowed only if the 100–200 curves show continuing multi-seed improvement without clinical-guardrail deterioration. A method- or seed-specific extension is not permitted.

Final and frozen-pretraining checkpoints are evaluated on four routing scenarios with at least 100 paired CRN replications per training seed; additional evaluation replication may narrow uncertainty without changing the trained policies. Lead-zero specimen transit and one-epoch product return are prespecified sensitivities. The primary attribution is GCN versus matched flat, GCN versus MDL-2, and final versus frozen pretraining. No routing is retained only for deterministic mechanics regression or a separately approved supplementary ablation, not as a default formal training arm. Development screens are reported as tuning evidence only, and no routing-primary headline claim is made until the untouched-seed outputs pass the locked gates.

5.3. *distributed manufacturing simulation*

To evaluate the proposed method, we now construct a simulation environment of a decentralized manufacturing system consisting of $N = 20$ manufacturing facilities and its dedicated reagent supplier. Each PRM product requires $T = 3$ epochs of manufacturing time, and the considered finite problem horizon contains $\Gamma = 52$ decision epochs. For every epoch t , there will be d_t^n new demand for facility n . The demand arrival process for facility

Algorithm 2 State normalization

Require: State variables: s_t^n , r_t^n , b_t^n , and p_t^n

1: Maximal values of the state variables for all facilities: M_s , M_r , M_b , and M_p

Ensure: Normalized state variables: $\hat{s}_t^n$, $\hat{r}_t^n$, $\hat{b}_t^n$, $\hat{p}_t^n$

2: **for** $n = 1$ to N **do**

3: $\hat{s}_t^n = s_t^n / M_s$

4: $\hat{r}_t^n = r_t^n / M_r$

5: $\hat{b}_t^n = b_t^n / M_b$

6: $\hat{p}_t^n = p_t^n / M_p$

7: **end for**

1008 n follows a Poisson distribution $\text{Poisson}(\lambda_n)$. The supplier disruption follows
1009 independent Bernoulli processes $\text{Ber}(\zeta_n)$, i.e., the facility n has a probability
1010 ζ_n to have supplier disruption and could not purchase any reagent for the
1011 epoch. This simulation can be further adjusted to reflect any real situation,
1012 e.g., supply disruption and demand uncertainty, that may be encountered.
1013 Such flexibility in using the simulation system will be demonstrated in Sec-
1014 tion 5.4 when we compare GCN-DDPG with other methods.

1015 Patient-condition parameters are calibrated before policy comparison.
1016 The nominal scenario uses $\lambda_H = 0.0013$, $\lambda_F = 0.0156$, Weibull shape $\kappa = 1.5$
1017 and scale $\theta = 4$, post-deterioration acceleration $\rho = 3$, eligibility threshold
1018 $\epsilon = 0.80$, and waiting-time pressure $0.0013\alpha^2$. We selected these values from
1019 a fixed-seed sweep whose pre-specified target was a 10–15% *patient ineligi-*
1020 *bility during manufacturing* rate over the three-week process. This outcome
1021 is not labeled a technical manufacturing failure: it records a patient who

1022 becomes clinically ineligible after production begins, whereas a process or
 1023 release-specification failure would be a separate stochastic event. The target
 1024 is consistent with the order of magnitude of pre-infusion attrition reported for
 1025 autologous CAR-T pathways [3] and with a 5% weekly eligibility-transition
 1026 benchmark, which gives $1 - (1 - 0.05)^3 = 14.26\%$ over three weeks.

1027 The patient-facing weights are $\omega_{\text{lost}} = 500,000$, $\omega_{\text{exp}} = 100,000$, and $\omega_{\text{urg}} =$
 1028 $25,000$. The earlier exploratory values (50,000, 40,000, 5,000) made one-time
 1029 patient loss cheaper than even one period of combined reagent and bioreactor
 1030 shortage, so a policy could lower its objective when a patient left the system.
 1031 The calibrated ordering removes this directional inconsistency and is held
 1032 fixed for all policy comparisons.

1033 The epoch is assumed to occur weekly. Reagent and idle-bioreactor trans-
 1034 fers enter an in-transit pipeline and arrive after one, two, or three epochs,
 1035 according to whether the geographic distance is at most 500 miles, between
 1036 500 and 1,500 miles, or above 1,500 miles. In the routing-primary campaign,
 1037 an intact waiting specimen may take one direct qualified route. The main
 1038 setting uses one transit epoch and immediate modeled return of finished
 1039 product to the collection facility; zero-epoch specimen transit and one-epoch
 1040 product return are sensitivities. The previously reported case studies set
 1041 $w_t^n = 0$ and are therefore not routing-effect experiments. After demand
 1042 and any due transfers or routes have arrived, the next observed state is
 1043 $\mathbf{x}_{t+1}^n = (s_{t+1}^n, r_{t+1}^n, \mathbf{b}^n, P_{t+1})$ for facility $n = 1, \dots, N$, according to Equation
 1044 (1). The state variables are updated by the following rules:

$$\begin{aligned}
s_{t+1}^n &= s_t^n - m_t^n + d_t^n + w_t^n \\
r_{t+1}^n &= r_t^n - m_t^n + e_t^n + p_t^n \\
b_{t+1}^{n,\tau} &= \begin{cases} b_t^{n,0} - m_t^n + b_t^{n,1} + q_t^n & \text{if } \tau = 0 \\ b_t^{n,\tau+1} & \text{if } 1 \leq \tau \leq T-2 \\ m_t^n & \text{if } \tau = T-1 \end{cases}
\end{aligned}$$

1045 where $m_t^n = \min(s_t^n, b_t^{n,0}, r_t^n)$ is the number of products that begin man-
1046 ufacturing at epoch t , d_t^n is the demand arriving at facility n in epoch t , and
1047 P_{t+1} follows an exogenous Markov chain from P_t . The operating component
1048 of the single-period objective can be decomposed as

$$\begin{aligned}
c_{\text{op}}(\mathbf{A}_t; \mathbf{X}_t) &= \sum_{n=1}^N C_{\text{op},t}^n, \\
C_{\text{op},t}^n &= C_{\text{Reg},t}^n + C_{\text{Bio},t}^n + C_{\text{Tran},t}^n.
\end{aligned}$$

1049 The meanings of $C_{\text{Reg},t}^n$, $C_{\text{Bio},t}^n$, and $C_{\text{Tran},t}^n$ are summarized below. These
1050 terms constitute only c_{op} ; the patient-related terms in Equation (7) must
1051 also be included to obtain the total weighted objective.

- 1052 • $C_{\text{Reg},t}^n = c_R p_t^n + h_R (r_{t+1}^n - s_{t+1}^n)^+ + p_R (s_{t+1}^n - r_{t+1}^n)^+$ is the cost related
1053 to reagent at epoch t , including the reagent replenishment cost $c_R p_t^n$,
1054 the reagent overstock holding cost $h_R (r_{t+1}^n - s_{t+1}^n)^+$, and the reagent
1055 understock penalty $p_R (s_{t+1}^n - r_{t+1}^n)^+$;
- 1056 • $C_{\text{Bio},t}^n = h_B (b_{t+1}^{n,0} - s_{t+1}^n)^+ + p_B (s_{t+1}^n - b_{t+1}^{n,0})^+$ is the cost related to

1057 bioreactor at epoch t , including the bioreactor overstock holding cost
1058 $h_B (b_{t+1}^{n,0} - s_{t+1}^n)^+$, and the bioreactor understock penalty $p_B (s_{t+1}^n - b_{t+1}^{n,0})^+$;
1059 • $C_{\text{Tran},t}^n = K_R |e_t^n| + K_B |q_t^n| + K_S |w_t^n|$ is the logistics-cost expression.
1060 The routing-primary campaign includes the patient-indexed specimen
1061 term; it vanishes only in the earlier no-routing case studies.

Table 7: Modeled cost coefficients used in the calibrated experiments. Values are weighted-objective units and are held fixed across policies. Transfer coefficients are base values before the geographic distance and travel-time surcharge. Patient-related coefficients are penalty or shadow-value parameters rather than observed cash expenditures.

Component	Coefficient	Interpretation
Reagent purchase, c_R	42,174.0	per purchased unit
Reagent holding, h_R	113.5	per excess unit-period
Reagent shortage, p_R	86,504.5	per shortage unit-period
Bioreactor holding, h_B	14.4	per idle unit-period
Bioreactor shortage, p_B	50,273.6	per shortage unit-period
Reagent transfer, K_R	200.0	per transferred unit, before surcharge
Capacity transfer, K_B	500.0	per relocated unit, before surcharge
Patient loss, ω_{lost}	500,000	per lost patient
Expiry, ω_{exp}	100,000	per wasted material or therapy unit
Urgency, ω_{urg}	25,000	per at-risk unserved patient-period

1062 5.4. Case Study 1: Interconnected Clinic Network Analysis

1063 This case study holds the demand prior equal to the arrival distribution
1064 and varies the supplier disruption rate over 0.05, 0.30, and 0.60. Its purpose

is to measure the value of graph-aware coordination when the deterministic heuristics are not disadvantaged by demand misspecification. The final matched multi-seed campaign for these three regimes is still pending; consequently, we do not report numerical disruption-effect claims in the present draft. This case study will use the same five training seeds, paired Monte Carlo evaluation, checkpoint selection, and two-level bootstrap protocol as Case Study 2.

5.5. Case Study 2: Adaptability to Non-Stationary Demand

This case study evaluates the main robustness hypothesis under patient-condition and geographic network dynamics. The 20 clinics are partitioned into four five-clinic demand groups. MDL-2 plans with Poisson prior rates $[7.5, 3.2, 6.0, 3.5]$, whereas the simulator generates arrivals using true rates $[10.5, 3.52, 7.5, 3.15]$. The environment also includes patient deterioration, expiry, clustered demand shocks, regional supplier disruptions, and distance-dependent transfer lead times. During training, demand rates are randomized between 0.8 and 1.5 of the scenario rates and forecast error between 0 and 0.35; evaluation is fixed to the scenario above.

Before retraining learned policies, we evaluated ISO and MDL-2 with 100 paired Monte Carlo replications to verify that the calibrated patient process and objective produce the intended clinical range. Table 8 reports this calibration pilot and the subsequent matched learned-policy screen. Learned-policy results produced before the complete patient lifecycle and objective calibration are not comparable and are therefore retired rather than carried forward into this experiment.

Both heuristics fall inside the pre-specified clinical calibration band: manufacturing-

Table 8: Calibrated demand-prior-drift results. ISO and MDL-2 use the initial 100-replication calibration stream; learned-policy and matched heuristic rows aggregate 100 paired replications for each of three training seeds. Objective values are the total modeled weighted objective in billions of cost units, not realized accounting expenditure. Lower objective, patient loss, and manufacturing-period ineligibility are better; higher completion service and eligibility are better.

Method	Objective (10^9 units)	Completion	Eligibility	Lost	Mfg. inelig.
Initial ISO calibration	3.990456	0.337612	0.377537	4021.92	0.146398
Initial MDL-2 calibration	3.895694	0.371948	0.421024	3698.87	0.121364
AFR-Flat-DDPG	3.927584	0.370359	0.507030	3730.36	0.122223
AFR-GCN-DDPG	3.927594	0.370359	0.507030	3730.35	0.122223
AFR-GCN-TD3	3.927628	0.370358	0.507030	3730.36	0.122223
MDL-2	3.927620	0.370359	0.507030	3730.36	0.122223
ISO	4.011547	0.337521	0.460044	4043.10	0.146468
pMYO	4.180485	0.343311	0.496776	3894.09	0.133332
MYO	4.194549	0.342889	0.496586	3897.01	0.133582

period patient ineligibility is 14.64% for ISO and 12.14% for MDL-2. MDL-2 reduces the mean total weighted objective by 94.76 million modeled cost units (2.37%) relative to ISO; the paired 95% interval for MDL-2 minus ISO is $[-108.81, -80.71]$ million cost units, and MDL-2 has a lower objective in 92 of 100 paired replications. It also loses 323.05 fewer patients on average, discards 49.40 fewer in-process therapies, and raises completion service by 0.03434.

The calibrated learned-policy screen used 100 training episodes and three training seeds. AFR-GCN-DDPG selected and deployed a nonzero residual in all three seeds; AFR-GCN-TD3 and AFR-Flat-DDPG each returned to MDL-2 in one seed. AFR-GCN-DDPG reduced the mean total weighted objective relative to MDL-2 by only 26,519 modeled cost units (0.000675%),

with a two-level bootstrap 95% interval of $[-89,588, 40,330]$ cost units. Its mean objective was 9,460 cost units higher than the matched flat residual policy, with interval $[-7,951, 24,621]$. Completion service and patient loss were effectively unchanged. Thus the screen demonstrates safeguard-compatible parity with a strong anchor, but it does not support either a material RL improvement or a graph-representation advantage. The accepted correction primarily substitutes additional reagent purchase and holding cost for lower reagent shortage cost. Because this residual changes replenishment only, the next matched experiment expands the bounded correction to distance-aware reagent and idle-bioreactor transfers, where geographic message passing can affect the decision.

The small headline percentage in this screen is intentionally computed against the complete weighted objective rather than a narrower, post hoc denominator. Patient-related penalties and capacity-shortage penalties can dominate that total, while a replenishment-only residual directly changes only a subset of the operating components. We therefore interpret 0.000675% as statistical parity rather than as a material economic saving. Subsequent network-residual comparisons report the total-objective change together with operating, bioreactor-shortage, transfer, and patient-penalty decompositions so that denominator dilution and the mechanism of any improvement remain visible.

5.5.1. Formal five-seed demand-prior-drift confirmation

The formal replenishment-residual comparison increased the training budget to 300 episodes and used five independent training seeds. Each selected checkpoint was evaluated on 100 paired holdout replications, yielding 500

1127 paired outcomes per method. Table 9 reports the mean outcomes, and Ta-
 1128 ble 10 reports paired differences. The learned-policy values in these tables
 1129 supersede the 100-episode diagnostic screen in Table 8.

Table 9: Five-seed demand-prior-drift confirmation. Objective values are the total modeled weighted objective in billions of units. Each row aggregates 100 paired holdout replications for each of five training seeds.

Method	Objective (10^9 units)	Completion	Eligibility	Waiting	Lost	At-risk unserved
MDL-2	1.268257	0.429982	0.530313	3.052625	3665.59	5206.41
AFR-Flat-DDPG	1.268089	0.430320	0.530523	3.052012	3663.28	5203.86
AFR-GCN-DDPG	1.268049	0.430321	0.530524	3.052012	3663.27	5203.86
AFR-GCN-TD3	1.268241	0.430317	0.530523	3.052019	3663.30	5203.88

Table 10: Paired five-seed demand-prior-drift comparisons. Differences are candidate minus baseline, so a negative objective difference favors the first method. Confidence intervals use the two-level bootstrap over training seeds and paired replications.

Comparison	ΔJ_{tot} (units)	Relative gap	Paired 95% CI	Wins / 500
AFR-GCN-DDPG vs. MDL-2	-207,485	-0.016360%	[-332,641, -91,367]	323
AFR-Flat-DDPG vs. MDL-2	-167,610	-0.013216%	[-286,899, -46,230]	319
AFR-GCN-DDPG vs. AFR-Flat-DDPG	-39,876	-0.003145%	[-89,702, 14,025]	285
AFR-GCN-TD3 vs. MDL-2	-15,823	-0.001248%	[-142,607, 99,437]	268
AFR-GCN-DDPG vs. AFR-GCN-TD3	-191,662	-0.015112%	[-234,659, -148,401]	377

1130 AFR-GCN-DDPG improves on MDL-2 in every training-seed mean and
 1131 has a two-level confidence interval fully below zero. Completion service in-
 1132 creases by 0.000340, eligibility by 0.000211, average waiting time falls by
 1133 0.000612, patients lost fall by 2.32, and at-risk unserved patients fall by 2.55.
 1134 This supports a statistically stable anchor improvement under demand-prior
 1135 drift, but not a large economic effect. The matched flat policy also improves
 1136 on MDL-2, and the direct graph-versus-flat interval includes zero. Replen-

1137 ishment residual learning is therefore supported in this scenario, whereas an
1138 independent graph-representation advantage is not.

1139 The matched TD3 policy preserves the patient-facing improvements but
1140 has a cost interval crossing zero. DDPG is lower cost than TD3 in all five
1141 training-seed means, and their paired interval is fully below zero. This
1142 comparison supports DDPG as the canonical backbone for the proposed
1143 replenishment-residual controller; it is not a claim that DDPG generally
1144 dominates TD3 outside this residual-learning design.

1145 *5.6. Case Study 3: Graph Attribution Under Regional Non-Stationarity*

1146 This earlier no-routing case study combines patient-condition dynamics,
1147 clinic geography, distance-based transfer delays, regional demand heterogene-
1148 ity, and demand regime changes. Unlike the replenishment-only experiment,
1149 the network residual contains a node head for signed replenishment correc-
1150 tions and edge heads for reagent and idle-bioreactor transfer corrections.
1151 Edge flows are projected to preserve resource conservation and feasibility,
1152 while patient-indexed specimen routing is disabled by that historical proto-
1153 col. This action space gives graph message passing a direct route to affect
1154 geographically structured decisions.

1155 *5.6.1. Network AFR-GCN-DDPG under an abrupt regional shift*

1156 The abrupt-shift policy uses transfer-focused advantage-filtered targets
1157 and a fixed deployment scale selected on 30 paired validation episodes. Three
1158 independently initialized graph policies and parameter-matched flat policies
1159 were then evaluated on the same 100 previously unseen paired holdout repli-
1160 cations per seed. The graph and flat models contain 524,775 and 528,404

trainable parameters, respectively, so the comparison does not give the graph policy a parameter-count advantage.

Table 11: Three-seed abrupt-regional-shift comparisons. Objective differences are in millions of modeled weighted-objective units. Each comparison contains 300 paired holdout outcomes.

Comparison	ΔJ_{tot} (10^6 units)	Relative gap	Paired 95% CI	Δ completion	Δ patients lost
Network AFR-GCN-DDPG vs. MDL-2	-25.39	-1.1337%	[-29.76, -20.95]	+0.001357	-0.70
Network AFR-GCN-DDPG vs. AFR-Flat-DDPG	-25.01	-1.1166%	[-29.35, -20.61]	+0.002836	-9.86

All three seed-level objective means favor the graph controller. Relative to MDL-2, the completion-service 95% interval is $[0.000240, 0.002458]$ and the patients-lost interval is $[-6.42, 5.07]$; aggregate patient-loss noninferiority therefore passes, but the strict per-seed clinical gate passes for two of three seeds. Relative to the matched flat policy, the completion interval is $[0.001766, 0.003898]$ and the patients-lost interval is $[-15.24, -4.31]$. This is the current statistically significant evidence for a graph-specific benefit: it appears when the action space includes geographically structured edge decisions, not when the actor modifies replenishment alone.

Table 12 explains why the total-objective percentage is smaller than the change in the component directly affected by the policy. MDL-2’s mean objective is 2.2399 billion units. Bioreactor shortage, patient loss, and expiry account for 44.10%, 43.53%, and 8.71%, respectively. The graph controller reduces the bioreactor-shortage component by 25.42 million units, or 2.57% of that component, which becomes a 1.13% improvement when divided by the complete objective. The component percentage is a mechanism measure; the complete-objective percentage remains the headline result.

This positive network checkpoint is driven primarily by advantage-filtered

Table 12: Cost decomposition for MDL-2 and Network AFR-GCN-DDPG in the abrupt-regional-shift holdout. The final column is the within-component percentage change. Components are grouped without changing the total-objective denominator used in Table 11.

Component	MDL-2 mean (10^6 units)	Share of total	AFR change (10^6 units)	Component change
Bioreactor shortage	987.891	44.10%	-25.420	-2.57%
Patient loss	975.120	43.53%	-0.352	-0.04%
Expiry	195.008	8.71%	-0.084	-0.04%
Urgency	6.118	0.27%	+0.641	+10.48%
Other operating components	75.742	3.38%	-0.178	-0.23%
Total weighted objective	2239.879	100.00%	-25.393	-1.13%

1181 teacher distillation. Earlier unconstrained offline DDPG and TD3 updates
 1182 weakened the policy or violated patient safeguards. The experiment therefore
 1183 establishes value from the graph-aware AFR controller, but it does not by
 1184 itself establish that online DDPG updates generated that value.

1185 5.6.2. Conservative online-TD3 attribution screen

1186 To test whether actual actor-critic updates add value beyond the frozen
 1187 pretrained controller, we initialized matched graph and flat residual TD3
 1188 policies from their corresponding pretrained checkpoints and ran 100 online
 1189 episodes for each of three seeds. Every run completed nonzero critic and actor
 1190 updates with finite diagnostics. Final and frozen checkpoints were evaluated
 1191 on identical 100-replication CRN holdouts.

Table 13: Final conservative TD3 outcomes under geographic regional demand drift. Values aggregate three training seeds and 100 paired holdout replications per seed.

Method	Objective (10^9 units)	Completion	Patients lost	Mfg. ineligibility
MDL-2	2.458807	0.490786	2211.86	0.073077
Final AFR-Flat-TD3	2.460346	0.490433	2213.75	0.073209
Final AFR-GCN-TD3	2.456263	0.491636	2206.70	0.072685

Table 14: Paired final-checkpoint TD3 comparisons. Objective differences are in millions of modeled weighted-objective units.

Comparison	ΔJ_{tot}	Relative gap	Paired 95% CI	Δ completion	Δ lost	Δ mfg. inelig.
Final AFR-GCN-TD3 vs. MDL-2	-2.544	-0.103459%	$[-3.872, -1.131]$	+0.000850	-5.16	-0.000392
Final AFR-Flat-TD3 vs. MDL-2	+1.539	+0.062595%	$[+0.807, +2.387]$	-0.000353	+1.89	+0.000132
Final AFR-GCN-TD3 vs. AFR-Flat-TD3	-4.083	-0.165951%	$[-5.633, -2.436]$	+0.001203	-7.05	-0.000524

The final graph policy improves on MDL-2 for each seed (-0.051936% , -0.139249% , and -0.119193%), deploys nonzero residuals in 73.50%, 47.63%, and 67.25% of evaluated decisions, and passes all pre-specified clinical non-inferiority checks. The matched flat policies are more expensive than MDL-2 and fail clinical noninferiority for all three seeds. The final graph-versus-flat comparison is therefore statistically and clinically favorable.

The online-learning attribution is more limited. The pooled final-minus-frozen graph difference is -0.350 million units, with 95% interval $[-1.032, 0.155]$ million, and the GCN-minus-flat difference-in-differences is -0.485 million units, with interval $[-1.489, 0.173]$ million. Both point estimates favor online TD3 updates, but both intervals include zero. Thus the final graph controller outperforms MDL-2 and matched flat TD3 in this regional screen, while the independent incremental benefit of online TD3 over frozen pretraining is not established. This earlier regional screen is retained as development context; the matched routing-primary TD3 ablation reported below supersedes it for the current routing evidence package.

5.6.3. Routing-primary five-seed confirmation

The final routing-primary experiment holds patient-indexed specimen routing admissible for every policy and therefore compares controllers rather than estimating a routing-versus-no-routing treatment effect. AFR-GCN-

DDPG and a parameter-matched AFR-Flat-DDPG controller were each trained for 100 online episodes with five independent seeds. Their fixed final and frozen-pretraining checkpoints were evaluated on four prespecified routing scenarios with 100 paired formal-holdout replications per training seed and scenario, yielding 2,000 paired outcomes per comparison. Table 15 reports candidate-minus-comparator effects.

Table 15: Formal routing-primary paired comparisons. Objective differences and intervals are in millions of modeled weighted-objective units. Negative cost, patients-lost, and manufacturing-ineligibility differences and positive completion differences favor the candidate.

Comparison	ΔJ_{tot}	Relative gap	Two-level 95% CI	Δ completion	Δ lost	Δ mfg. inelig.
AFR-GCN-DDPG vs. MDL-2	-17.690	-0.658%	[-19.361, -15.725]	+0.001806	-25.263	-0.004705
AFR-Flat-DDPG vs. MDL-2	-9.086	-0.338%	[-10.587, -7.510]	+0.000636	-8.888	-0.003773
AFR-GCN-DDPG vs. AFR-Flat-DDPG	-8.604	-0.321%	[-11.065, -6.428]	+0.001169	-16.375	-0.000932

AFR-GCN-DDPG is lower cost than MDL-2 in all 20 training-seed-by-scenario cells. Its two-level intervals are wholly favorable against both MDL-2 and the matched flat controller for total cost, completion service, patients lost, and manufacturing ineligibility. Figure 3 places the formal comparisons beside the prespecified transport-timing sensitivity and the separately labeled TD3 development ablation.

The MDL-2 mean objective is 2,686.444 million units. Its additive components are base operating cost (45.856%), patient-loss cost (44.841%), expiry cost (8.966%), and urgency cost (0.337%). The GCN improvement decomposes into -2.100, -12.631, -2.806, and -0.152 million units for these four components, respectively. Within base operating cost, specimen-transfer cost increases by 0.174 million while the remaining operating and shortage sub-components decrease by 2.275 million. Specimen transfer is therefore a mech-

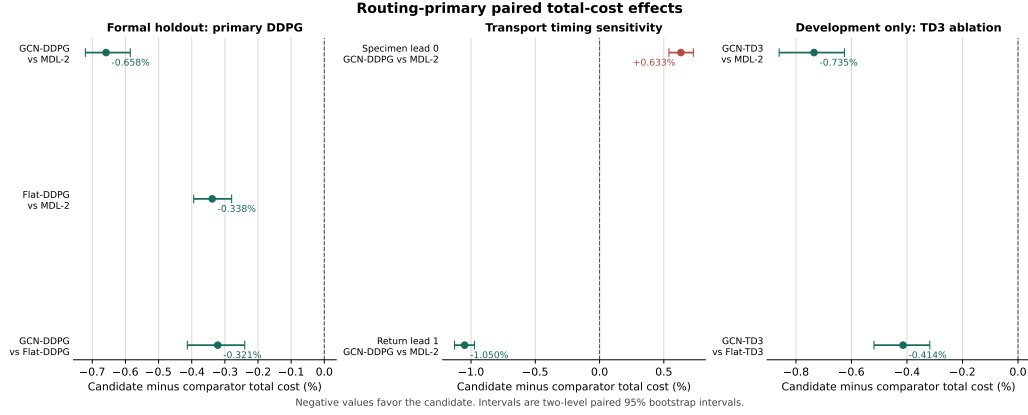


Figure 3: Routing-primary paired total-cost effects. Negative values favor the candidate, and bars are two-level paired 95% bootstrap intervals. The left panel is formal holdout evidence. The center panel is frozen-policy transport-timing sensitivity. The right panel is a development-only TD3 backbone ablation and is not pooled with the formal result.

anistic subcomponent of base cost and is not added to the four-component total a second time. The dominant benefit is lower modeled patient-loss cost, not direct transportation savings. Figure 4 keeps the additive objective decomposition separate from its base-cost detail.

The transport-timing result is asymmetric rather than uniformly robust. When specimens become available immediately (lead 0 rather than lead 1), GCN cost is 0.633% higher than MDL-2 and 0.255% higher than flat. With a one-epoch finished-product return delay, GCN is 1.050% lower than MDL-2 and 0.303% lower than flat. These are epoch-level availability assumptions, distinct from continuous geographic travel time, and the reversal precludes a blanket transport-timing robustness claim.

Finally, the formal final-minus-frozen GCN point estimate is +0.073 million units (+0.00273%), indicating no material online increment. On the

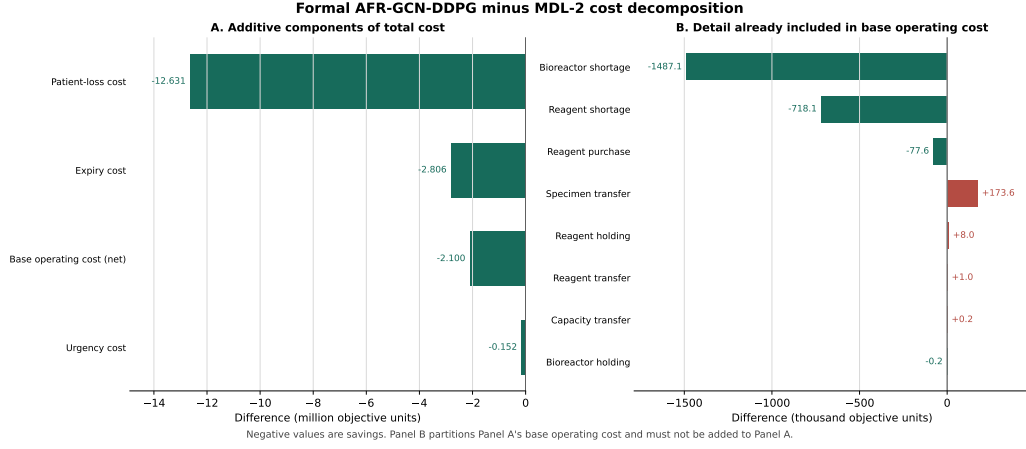


Figure 4: Formal AFR-GCN-DDPG minus MDL-2 cost decomposition. Negative values are savings. Panel A contains the four components that add to total cost. Panel B partitions panel A’s base operating cost; it is shown for mechanism interpretation and must not be added to panel A.

1244 separate paired development checkpoint curve, episode 100 minus episode 75
1245 is $+0.069$ million with interval $[-0.345, +0.502]$ million. Howard’s matched
1246 three-seed TD3 development ablation corroborates graph and anchor value:
1247 final GCN-TD3 is 0.735% lower cost than MDL-2 and 0.414% lower than flat
1248 TD3. However, final GCN-TD3 is 0.00080% worse than its frozen-pretraining
1249 checkpoint, and its audited paired interval crosses zero. A subsequent struc-
1250 tured specimen-option exploration screen achieved the intended action cover-
1251 age but produced a GCN final-minus-frozen effect of $+0.00416\%$ with inter-
1252 val $[-0.02766, +0.04091]\%$. Thus neither actor-critic backbone establishes
1253 an independent online-learning gain at the tested budget.

1254 *5.6.4. Prospective online-DDPG attribution and action-geometry audits*

1255 The final-versus-frozen comparisons above motivated a prospective DDPG
1256 attribution experiment rather than another unconstrained tuning cycle. Stage
1257 F1 created tensor-identical control and candidate forks with the same struc-
1258 tured specimen exploration. The sole candidate difference was an auxiliary
1259 counterfactual-advantage critic loss computed from exact within-state, four-
1260 step behavior-versus-MDL-2 rollouts. All 12 training jobs completed 100
1261 online episodes, and the two prespecified checkpoint curves contained 3,600
1262 learned and 3,600 anchor outcomes in total.

1263 For the primary GCN arm, candidate final minus its frozen pretraining
1264 checkpoint was $-21,298$ modeled objective units (-0.001067%), with paired
1265 95% interval $[-715,371, +527,375]$. Only one of three seeds was favorable, so
1266 the prospective two-seed and wholly-below-zero gate failed. Candidate final
1267 minus matched control final was $+4,081$ units ($+0.000205\%$), with interval
1268 $[-95,056, +95,036]$ and 131 exact ties among 150 pairs. The auxiliary target
1269 was sampled and partly learned (persisted sign accuracy 0.709–0.786), but
1270 contributed only 0.245–0.343% of total critic loss and did not produce a
1271 materially different deployed policy.

1272 Read-only transfer audits then examined why additional critic structure
1273 did not reach execution. In the nominal-history G0 diagnostic, the candidate
1274 final critic selected the independently rolled-out best remaining-horizon legal
1275 action in 3/27 states, versus 2/27 for control; only 2/27 raw actor differences
1276 survived correction composition, projection, and integer-lot execution. An
1277 all-legal-action, remaining-horizon ranker in G1 reached 47/156 held-out-seed
1278 top-1 decisions, versus 46/156 for the frozen critic. The original F0/G0/G1

1279 scenario-specific labels were subsequently corrected because those post-hoc
 1280 audits had reconstructed nominal history rather than the assigned online
 1281 scenario. Their immutable rows remain useful mechanism diagnostics, but
 1282 no persistent-hotspot or per-cluster inference is made from them.

1283 Corrected prospective action-geometry screens reconstructed the benchmark-
 1284 plan scenarios directly. Table 16 reports the fraction of audited states with an
 1285 independently validated action saving at least one million modeled objective
 1286 units without worsening completion, patients lost, or manufacturing ineligi-
 1287 bility. Specimen routing retained local headroom, whereas neither continuous
 1288 resource group produced a prospectively selected material improvement.

Table 16: Corrected frozen action-geometry screen. H0 is the persistent-shift screen and H1 is the nonstationary screen. “Opp.” is the fraction of 27 audited states with a material clinically noninferior action; “selected” applies the same test to the action chosen on an independent discovery stream.

Action group	H0 opp.	H0 selected	H1 opp.	H1 selected
Specimen transfer	55.6%	40.7%	55.6%	37.0%
Reagent transfer	3.7%	0.0%	3.7%	0.0%
Reagent plus capacity	3.7%	0.0%	0.0%	0.0%

1289 Together, these audits narrow rather than erase the reinforcement-learning
 1290 contribution. The learned graph residual policy is supported by the formal
 1291 policy comparisons and the TD3 backbone ablation. What is not supported
 1292 is a separate claim that the tested online Bellman and policy-gradient up-
 1293 dates created the endpoint gain. In the current simulator, useful local control
 1294 authority is concentrated in projected discrete specimen choices, while the
 1295 channels that preserve continuous action geometry expose little clinically

1296 noninferior material headroom.

1297 **6. Discussion and Conclusions**

1298 This study develops a patient-condition and geography-aware simulation
1299 of a 20-clinic PRM network, including identity-preserving pre-manufacturing
1300 specimen routing, and a residual GCN-DDPG controller that learns around a
1301 strong MDL-2 anchor. Advantage-filtered distillation identifies sparse state-
1302 dependent corrections, while checkpoint selection, residual-scale calibration,
1303 and fallback prevent an unsupported correction from being deployed. This
1304 structure preserves the interpretability of the anchor while allowing the graph
1305 actor to respond to patient pressure, transfer pipelines, regional disruptions,
1306 and demand-prior drift.

1307 The routing-primary formal result now supplies the main controller ev-
1308 idence. With routing held admissible for all policies, AFR-GCN-DDPG
1309 improves on both routing MDL-2 and a parameter-matched flat controller,
1310 and all reported patient-facing metrics move in the favorable direction. The
1311 graph-versus-flat interval is wholly below zero, so this matched experiment
1312 supports a graph-representation advantage rather than only generic residual
1313 learning. Routing itself is not the treatment in this comparison; estimating
1314 the value of making routing admissible would require a separately locked
1315 routing-versus- no-routing design.

1316 The component analysis also changes the interpretation of the result.
1317 Lower patient-loss cost accounts for most of the total reduction, followed by
1318 expiry and net operating cost. Specimen-transfer cost rises, as expected when
1319 a graph controller makes greater use of geographic reallocation, but this in-

crease is more than offset by reductions elsewhere. Reporting the component changes alongside the full-objective percentage preserves the preregistered denominator and shows that the gain is patient- and shortage-facing rather than an artifact of inexpensive transport.

The method attribution is narrower than the policy comparison. Frozen advantage-filtered pretraining already captures nearly all observed DDPG and TD3 value. The prospective F1 paired-critic experiment also failed its final-versus-frozen and candidate-versus-control gates. Mechanism audits found weak long-horizon legal-action ranking, severe attenuation through projection and integer-lot execution, and little clinically noninferior headroom in the simulator’s genuinely continuous resource channels. We therefore distinguish three supported layers: the constrained graph-MDP formulation, the advantage-filtered graph residual policy class, and its calibrated execution pipeline. The formal evidence supports the complete learned controller and a graph-versus-flat advantage; it does not establish an independent contribution from the tested online actor-critic updates. DDPG remains the canonical backbone because it is the prespecified five-seed formal method. The matched TD3 result is a useful development ablation, not evidence that DDPG generally dominates TD3 or that TD3 has formal-holdout status.

Several limitations remain. The routing-primary DDPG comparison has five training seeds, but the matched TD3 and mechanism-diagnostic screens have only three and use development streams. The formal comparison isolates graph versus flat representation and policy versus MDL-2, but it does not isolate routing availability or the incremental contribution of online updates. F1 and the corrected H0/H1 action-geometry screens use three train-

1345 ing seeds and bounded state samples; they reject their prespecified advance-
1346 ment gates but do not prove that every possible online algorithm or contin-
1347 uous action is ineffective. The original F0/G0/G1 post-hoc rows reconstruct
1348 nominal history, so their former scenario-specific labels are superseded and
1349 only their mechanism-level diagnostics are retained. Transport timing is not
1350 uniformly robust, and continuous travel-time viability is not coupled to the
1351 epoch-level availability sensitivity. Pure-policy, SAC, and PPO controllers
1352 have not been evaluated under the same locked routing-primary budget and
1353 should not be ranked from these data. The clinical calibration target is
1354 literature aligned but is not fitted to patient-level data from a single indi-
1355 cation, so sensitivity bands remain necessary. Patient-loss, expiry, urgency,
1356 and shortage coefficients are modeled decision weights rather than a complete
1357 accounting valuation; external economic and clinical calibration is required
1358 before interpreting objective differences as realized monetary savings or de-
1359 ploying the controller operationally. The routing-capable environment has
1360 passed implementation-level identity, conservation, lifecycle, and CRN tests,
1361 and the five-seed routing-primary GCN/flat confirmation is complete. Its for-
1362 mal claim is about controllers under a common routing-capable environment,
1363 not about the causal value of routing. Transport-timing sensitivity is asym-
1364 metric, with the primary advantage reversing under immediate specimen
1365 availability, so general timing robustness is not claimed. Development-only
1366 TD3 and DDPG mechanism screens remain labeled and archived separately
1367 from formal holdout evidence.

1368 **Future Work** This study motivates several extensions. Patient condi-
1369 tion can be represented with longitudinal clinical covariates rather than the

1370 current eligibility and survival-risk summary. Raw-material quality, product
1371 quality, process failures, machine breakdowns, and operator availability can
1372 be added to the simulator. The completed routing-primary protocol and its
1373 evidence map should now remain frozen. Future empirical work should use
1374 new research questions and independent streams rather than repeatedly tun-
1375 ing online attribution on the existing development evidence. Valuable exten-
1376 sions include unseen clinic networks, continuous transport-viability coupling,
1377 transfer disruptions, externally calibrated economic weights, and a separately
1378 prespecified routing-versus-no- routing study. If online deterministic policy
1379 gradients are to be made central, a new model should expose stable con-
1380 tinuous operational authority through decisions such as overtime, staffing,
1381 production-rate, procurement, or continuous capacity reservation, with new
1382 comparators and independent streams. Longer-term models should integrate
1383 production scheduling, product preparation and distribution, and treatment
1384 operations.

1385 **Data Availability**

1386 All study data are generated by the simulation model. The accompany-
1387 ing repository contains the source code, locked experiment configurations,
1388 curated summary outputs, training diagnostics, artifact hashes, and a claim-
1389 to-evidence map. Large checkpoint tensors and row-level Monte Carlo files
1390 are omitted from version control but are SHA256-inventoried and available
1391 from the corresponding author upon reasonable request.

1392 Declaration of Competing Interest

1393 The author declares that there is no known competing financial interest or
1394 personal relationship that could have appeared to influence the work reported
1395 in this paper.

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